

Example 1: Find the first order partial derivatives of the following

(i) $u = xy^2$ (ii) $u = \frac{x}{x^2 + y^2}$ (iii) $v = \tan^{-1}\left(\frac{x}{y}\right)$ (iv) $u = 3x^2y - x \sin xy$.

Solution:

(i) Given that $u = xy^2$

$$\frac{\partial u}{\partial x} = y^2 (1) \quad (\text{treating } y \text{ as constant}).$$

$$\frac{\partial u}{\partial y} = x(2y) = 2xy \quad (\text{treating } x \text{ as constant}).$$

(ii) Given that $u = \frac{x}{x^2 + y^2}$ $D\left(\frac{u}{v}\right) = \frac{vu' - uv'}{v^2}$

$$\frac{\partial u}{\partial x} = \frac{(x^2 + y^2)(1) - x(2x + 0)}{(x^2 + y^2)^2} = \frac{x^2 + y^2 - 2x^2}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2}$$

$$\frac{\partial u}{\partial y} = \frac{(x^2 + y^2)(0) - x(0 + 2y)}{(x^2 + y^2)^2} = \frac{0 - 2xy}{(x^2 + y^2)^2} = \frac{-2xy}{(x^2 + y^2)^2}$$

(iii) Given that $v = \tan^{-1}\left(\frac{x}{y}\right)$

$$\frac{\partial v}{\partial x} = \frac{1}{y} \cdot \frac{1}{1 + \left(\frac{x}{y}\right)^2} = \left(\frac{y}{y^2 + x^2}\right), \quad \frac{\partial v}{\partial y} = \frac{1}{1 + \left(\frac{x}{y}\right)^2} \cdot \frac{-x}{y^2} = \frac{-x}{y^2 + x^2}$$

(iv) Given that $u = 3x^2y - x \sin xy$... (1)

Differentiating equation (1) partially with respect to 'x' we get,

$$\begin{aligned} \frac{\partial u}{\partial x} &= 3(2x)y - \{x(\cos xy)y + \sin xy (1)\} \\ &= 6xy - xy \cos xy - \sin xy \end{aligned}$$

Differentiating equation (1) partially with respect to 'y' we get

$$\frac{\partial u}{\partial y} = 3x^2(1) - x \cos xy(x) = 3x^2 - x^2 \cos xy$$

Example 2: If $u = (1 - 2xy + y^2)^{-\frac{1}{2}}$, prove that $x \frac{\partial u}{\partial x} - y \frac{\partial u}{\partial y} = y^2 u^3$

Solution: Given that $u = (1 - 2xy + y^2)^{-\frac{1}{2}}$

$$\frac{\partial u}{\partial x} = -\frac{1}{2}(1 - 2xy + y^2)^{-\frac{1}{2}-1}(-2y) = \frac{y}{(1 - 2xy + y^2)^{\frac{3}{2}}}$$

$$\frac{\partial u}{\partial y} = \frac{-1}{2}(1 - 2xy + y^2)^{-\frac{1}{2}-1}(-2x + 2y) = \frac{x - y}{(1 - 2xy + y^2)^{\frac{3}{2}}}$$

$$\therefore x \frac{\partial u}{\partial x} - y \frac{\partial u}{\partial y} = \frac{xy}{(1 - 2xy + y^2)^{\frac{3}{2}}} - \frac{(xy - y^2)}{(1 - 2xy + y^2)^{\frac{3}{2}}} y^2 (1 - 2xy + y^2)^{-\frac{3}{2}}$$

$$x \frac{\partial u}{\partial x} - y \frac{\partial u}{\partial y} = y^2 u^3 \quad \left(\because u^3 = (1 - 2xy + y^2)^{-\frac{3}{2}} \right)$$

Hence proved.

Example 3: Find the first order partial derivative of $u = \tan^{-1} \frac{x^2 + y^2}{x + y}$

Solution:

$$\text{Given that } u = \tan^{-1} \frac{x^2 + y^2}{x + y} \quad \dots (1)$$

Differentiating equation (1) partially with respect to 'x', we get

$$\frac{\partial u}{\partial x} = \frac{1}{1 + \left(\frac{x^2 + y^2}{x + y} \right)^2} \cdot \frac{\partial}{\partial x} \left(\frac{x^2 + y^2}{x + y} \right)$$

$$\begin{aligned}
 &= \frac{1}{1 + \left(\frac{x^2 + y^2}{x + y}\right)^2} \left[\frac{(x + y) \left(\frac{\partial}{\partial x} (x^2 + y^2) \right) - (x^2 + y^2) \frac{\partial}{\partial x} (x + y)}{(x + y)^2} \right] \\
 &= \frac{(x + y)^2}{(x + y)^2 + (x^2 + y^2)^2} \left[\frac{(x + y)(2x) - (x^2 + y^2)(1)}{(x + y)^2} \right] \\
 &= \frac{(x + y)(2x) - (x^2 + y^2)}{(x + y)^2 + (x^2 + y^2)^2} = \frac{2x^2 + 2xy - x^2 - y^2}{(x + y)^2 + (x^2 + y^2)^2} \\
 \frac{\partial u}{\partial x} &= \frac{x^2 + 2xy - y^2}{(x + y)^2 + (x^2 + y^2)^2}
 \end{aligned}$$

Differentiating equation (1) partially with respect to ‘ y ’, we get

$$\begin{aligned}
 \frac{\partial u}{\partial y} &= \frac{1}{1 + \left(\frac{x^2 + y^2}{x + y}\right)^2} \cdot \left[\frac{(x + y)(2y) - (x^2 + y^2)(1)}{(x + y)^2} \right] \\
 &= \frac{(x + y)^2}{(x + y)^2 + (x^2 + y^2)^2} \left[\frac{2xy + 2y^2 - x^2 - y^2}{(x + y)^2} \right] \\
 \therefore \frac{\partial u}{\partial y} &= \frac{y^2 + 2xy - x^2}{(x + y)^2 + (x^2 + y^2)^2}
 \end{aligned}$$

Example 4: If $u = \log(\tan x + \tan y)$, **prove that** $\sin 2x \frac{\partial u}{\partial x} + \sin 2y \frac{\partial u}{\partial y} = 2$

Solution:

Given that $u = \log(\tan x + \tan y)$... (1)

Differentiating equation (1) partially with respect to ‘ x ’, we get

$$\frac{\partial u}{\partial x} = \frac{1}{\tan x + \tan y} \cdot \frac{\partial}{\partial x} (\tan x + \tan y)$$

$$\therefore \frac{\partial u}{\partial x} = \frac{\sec^2 x}{\tan x + \tan y}$$

Differentiating equation (1) partially with respect to 'y', we get

$$\begin{aligned} \frac{\partial u}{\partial y} &= \frac{1}{\tan x + \tan y} \cdot \frac{\partial}{\partial y} (\tan x + \tan y) \\ &= \frac{\sec^2 y}{\tan x + \tan y} \end{aligned}$$

$$\begin{aligned} \text{L.H.S} &= \sin 2x \frac{\partial u}{\partial x} + \sin 2y \frac{\partial u}{\partial y} \\ &= \frac{1}{\tan x + \tan y} [\sin 2x \sec^2 x + \sin 2y \sec^2 y] \\ &= \frac{1}{\tan x + \tan y} \left[2 \sin x \cos x \frac{1}{\cos^2 x} + 2 \sin y \cos y \cdot \frac{1}{\cos^2 y} \right] \\ &= \frac{1}{\tan x + \tan y} \left[2 \frac{\sin x}{\cos x} + 2 \frac{\sin y}{\cos y} \right] = \frac{2(\tan x + \tan y)}{(\tan x + \tan y)} \\ &= 2 = \text{R.H.S.} \end{aligned}$$

Example 9: If $z = \log(e^x + e^y)$, Show that $rt - s^2 = 0$ where $r = \frac{\partial^2 z}{\partial x^2}$; $t = \frac{\partial^2 z}{\partial y^2}$;

$$s = \frac{\partial^2 z}{\partial x \partial y}.$$

Solution:

Given that $z = \log(e^x + e^y)$

$$\frac{\partial z}{\partial x} = \frac{1}{e^x + e^y} \cdot e^x = \frac{e^x}{e^x + e^y}; \quad \frac{\partial z}{\partial y} = \frac{1}{e^x + e^y} \cdot e^y$$

$$r = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right) = \frac{(e^x + e^y)(e^x) - e^x(e^x)}{(e^x + e^y)^2} = \frac{(e^x)^2 + e^x e^y - (e^x)^2}{(e^x + e^y)^2}$$

$$r = \frac{e^{x+y}}{(e^x + e^y)^2}$$

By symmetry, $t = \frac{\partial^2 z}{\partial y^2} = \frac{e^{x+y}}{(e^x + e^y)^2} = r$

$$\begin{aligned} s &= \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right) = \frac{\partial}{\partial x} \left[\frac{e^y}{e^x + e^y} \right] \\ &= \frac{(e^x + e^y)(0) - e^y(e^x)}{(e^x + e^y)^2} = \frac{-(e^y)(e^x)}{(e^x + e^y)^2} \end{aligned}$$

$$s = \frac{-e^{x+y}}{(e^x + e^y)^2} = -r$$

$$rt - s^2 = r(r) - (-r)^2 = r^2 - r^2 = 0 \quad .$$

Example 10: If $v = (x^2 + y^2 + z^2)^{-1/2}$, show that $\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = 0$

Solution:

Given that $v = (x^2 + y^2 + z^2)^{-1/2}$

$$\frac{\partial v}{\partial x} = \frac{-1}{2} (x^2 + y^2 + z^2)^{-3/2} \cdot (2x) = -x (x^2 + y^2 + z^2)^{-3/2}$$

$$\frac{\partial^2 v}{\partial x^2} = - \left[x \cdot \frac{-3}{2} (x^2 + y^2 + z^2)^{-5/2} (2x) + (x^2 + y^2 + z^2)^{-3/2} (1) \right]$$

$$\frac{\partial^2 v}{\partial x^2} = 3x^2 (x^2 + y^2 + z^2)^{-5/2} - (x^2 + y^2 + z^2)^{-3/2}$$

Similarly, $\frac{\partial^2 v}{\partial y^2} = 3y^2 (x^2 + y^2 + z^2)^{-5/2} - (x^2 + y^2 + z^2)^{-3/2}$

$$\frac{\partial^2 v}{\partial z^2} = 3z^2 (x^2 + y^2 + z^2)^{-5/2} - (x^2 + y^2 + z^2)^{-3/2}$$

$$\begin{aligned} \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} &= 3x^2 (x^2 + y^2 + z^2)^{-5/2} - (x^2 + y^2 + z^2)^{-3/2} \\ &\quad + 3y^2 (x^2 + y^2 + z^2)^{-5/2} - (x^2 + y^2 + z^2)^{-3/2} \\ &\quad + 3z^2 (x^2 + y^2 + z^2)^{-5/2} - (x^2 + y^2 + z^2)^{-3/2} \\ &= 3(x^2 + y^2 + z^2)^{-5/2} (x^2 + y^2 + z^2) - 3(x^2 + y^2 + z^2)^{-3/2} \\ &= 3(x^2 + y^2 + z^2)^{-3/2} - 3(x^2 + y^2 + z^2)^{-3/2} = 0 \text{ Hence proved.} \end{aligned}$$

2.5 HOMOGENEOUS FUNCTIONS

A function $f(x, y, z)$ is called a homogeneous function of degree n if $f(tx, ty, tz) = t^n f(x, y, z)$. Here n need not be an integer. (i.e.) Replacing x, y, z by tx, ty, tz in the function $f(x, y, z)$, it reduces to $t^n f(x, y, z)$.

For example, if $f(x, y, z) = ax^2y + 2hxyz^2 + by^2z$

$$\begin{aligned} f(tx, ty, tz) &= a(tx)^2(yt) + 2h(tx)(zt)^2 + b(yt)^2(zt) \\ &= ax^2yt^3 + 2hxyz^2t^3 + by^2zt^3 \\ &= t^3(ax^2y + 2hxyz^2 + by^2z) = t^3 f(x, y, z) \end{aligned}$$

Hence the function $f(x, y, z)$ is homogeneous of degree 3.

2.5.1. Euler's Theorem on Homogeneous Functions

If $f(x, y)$ is a homogeneous function of degree n in x and y , then $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = nf(x, y)$

Proof:

$$\text{Let } f(x, y) = x^n g\left(\frac{y}{x}\right) \quad \dots (1)$$

Also $g'\left(\frac{y}{x}\right)$ denotes the derivative of $g\left(\frac{y}{x}\right)$ with respect to $\frac{y}{x}$.

Differentiating equation (1) partially with respect to x , we have

$$\begin{aligned} \frac{\partial f}{\partial x} &= x^n g'\left(\frac{y}{x}\right) \left(\frac{-y}{x^2}\right) + nx^{n-1} g\left(\frac{y}{x}\right) \\ \frac{\partial f}{\partial x} &= -yx^{n-2} g'\left(\frac{y}{x}\right) + nx^{n-1} g\left(\frac{y}{x}\right) \end{aligned}$$

Multiplying by x on both sides,

$$x \frac{\partial f}{\partial x} = nx^n g\left(\frac{y}{x}\right) - yx^{n-1} g'\left(\frac{y}{x}\right) \quad \dots (2)$$

Differentiating equation (1) partially with respect to y , we have

$$\begin{aligned} \frac{\partial f}{\partial y} &= x^n g'\left(\frac{y}{x}\right) \cdot \left(\frac{1}{x}\right) = x^{n-1} g'\left(\frac{y}{x}\right) \\ y \frac{\partial f}{\partial y} &= yx^{n-1} g'\left(\frac{y}{x}\right) \quad \dots (3) \end{aligned}$$

Adding equations (2) & (3) gives, $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = nx^n g\left(\frac{y}{x}\right)$

Using (1), $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = nf(x, y)$.

Note: This result can be extended for several independent variables as follows:

If, $f(x, y, z, \dots)$ is homogeneous function of the n^{th} degree in the variable $x, y, z, \dots$ then we

$$\text{get } \left(x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} + z \frac{\partial f}{\partial z} + \dots \right) = n f(x, y, z, \dots)$$

Points to Remember:

1. A function $f(x, y, z)$ is called a homogeneous function of degree n if $f(tx, ty, tz) = t^n f(x, y, z)$.

2. Euler's Theorem: If $f(x, y)$ is a homogeneous function of degree n in x and y ,

$$\text{then } x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = n f(x, y)$$

Example 1: Verify Euler's theorem for $f(x, y, z) = 3x^2yz + 5xy^2z + 4z^4$

Solution:

Given that $f(x, y, z) = 3x^2yz + 5xy^2z + 4z^4$

$$\frac{\partial f}{\partial x} = f_x = 6xyz + 5y^2z$$

$$\frac{\partial f}{\partial y} = f_y = 3x^2z + 10xyz$$

$$\frac{\partial f}{\partial z} = f_z = 3x^2y + 5xy^2 + 16z^3$$

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} + z \frac{\partial f}{\partial z} = 6x^2yz + 5xy^2z + 3x^2zy + 10xy^2z + 3x^2yz + 5xy^2z + 16z^4$$

$$= 12x^2yz + 20xy^2z + 16z^4 = 4(3x^2yz + 5xy^2z + 4z^4)$$

$$= 4f(x, y, z).$$

Example 2: If $u = \sin^{-1} \left(\frac{\sqrt{x} - \sqrt{y}}{\sqrt{x} + \sqrt{y}} \right)$ show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 0$

Solution:

Given that $u = \sin^{-1} \left(\frac{\sqrt{x} - \sqrt{y}}{\sqrt{x} + \sqrt{y}} \right)$

$$\sin u = \frac{\sqrt{x} - \sqrt{y}}{\sqrt{x} + \sqrt{y}} = \frac{\sqrt{x} \left(1 - \frac{\sqrt{y}}{\sqrt{x}} \right)}{\sqrt{x} \left(1 + \frac{\sqrt{y}}{\sqrt{x}} \right)} = \frac{\left(1 - \frac{\sqrt{y}}{\sqrt{x}} \right)}{\left(1 + \frac{\sqrt{y}}{\sqrt{x}} \right)}$$

Let $f = \sin u$, it is a homogeneous function of degree 0 in x, y .

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = 0 \cdot f$$

$$\text{(i.e.) } x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = 0 \quad \dots (1)$$

$$f = \sin u$$

$$\frac{\partial f}{\partial x} = \cos u \frac{\partial u}{\partial x}; \quad \frac{\partial f}{\partial y} = \cos u \frac{\partial u}{\partial y}$$

Substituting in equation (1) we get,

$$x \cos u \frac{\partial u}{\partial x} + y \cos u \frac{\partial u}{\partial y} = 0$$

$$\text{Dividing by } \cos u, \text{ we get } x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 0$$

Example 3: If $u = \cos \left(\frac{xy + yz + zx}{x^2 + y^2 + z^2} \right)$, Prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$.

Solution:

$$\text{Let } f(x, y, z) = u = \cos \left(\frac{xy + yz + zx}{x^2 + y^2 + z^2} \right)$$

$$\begin{aligned}
 f(tx, ty, tz) &= \cos \left(\frac{xyt^2 + yzt^2 + zxt^2}{x^2t^2 + y^2t^2 + z^2t^2} \right) \\
 &= t^0 \cos \left(\frac{xy + yz + zx}{x^2 + y^2 + z^2} \right) \\
 &= t^0 f(x, y, z)
 \end{aligned}$$

$\therefore u$ is a homogeneous function of degree zero.

Example 4: If $u = \tan^{-1} \left(\frac{x^3 + y^3}{x + y} \right)$, Prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$.

Solution:

Given that $u = \tan^{-1} \frac{x^3 + y^3}{x + y}$

$$\tan u = \frac{x^3 + y^3}{x + y}$$

Let $f(x, y) = \tan u = \frac{x^3 + y^3}{x + y}$

Hence f is a homogeneous function of degree 2.

By Euler's theorem, $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = 2f$

$$x \frac{\partial (\tan u)}{\partial x} + y \frac{\partial (\tan u)}{\partial y} = 2 \tan u$$

$$x \sec^2 u \frac{\partial u}{\partial x} + y \sec^2 u \frac{\partial u}{\partial y} = 2 \tan u$$

Dividing by $\sec^2 u$, we get $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{2 \tan u}{\sec^2 u}$

$$= 2 \frac{\sin u}{\cos u} \times \cos^2 u = 2 \sin u \cos u = \sin 2u$$

$$\therefore x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$$

Example 5: If $z = \sin^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{y}{x}\right) + \left(\frac{y}{x}\right)^2 + \sqrt{\frac{y}{x}}$, prove that $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 0$

Solution:

$$\text{Given that } z = f(x, y) = \sin^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{y}{x}\right) + \left(\frac{y}{x}\right)^2 + \sqrt{\frac{y}{x}}$$

$$\begin{aligned} f(tx, ty) &= \sin^{-1}\left(\frac{tx}{ty}\right) + \tan^{-1}\left(\frac{ty}{tx}\right) + \left(\frac{ty}{tx}\right)^2 + \sqrt{\frac{ty}{tx}} \\ &= t^0 \left[\sin^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{y}{x}\right) + \left(\frac{y}{x}\right)^2 + \sqrt{\frac{y}{x}} \right] \end{aligned}$$

$$f(tx, ty) = t^0 f(x, y)$$

$\therefore z = f(x, y)$ is homogeneous function of degree zero.

By Euler's theorem, $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 0(z) = 0$.

Example 6: If $z = x f\left(\frac{y}{x}\right) + \phi\left(\frac{x}{y}\right)$, prove that $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = x f\left(\frac{y}{x}\right)$.

Solution:

$$\text{Let } z = f_1(x, y) + f_2(x, y)$$

$\therefore f_1(x, y) = x f\left(\frac{y}{x}\right)$ is homogeneous of degree 1.

$f_2(x, y) = \phi\left(\frac{x}{y}\right)$ is homogeneous of degree zero.

By Euler's theorem,

$$\therefore x \frac{\partial f_1}{\partial x} + y \frac{\partial f_1}{\partial y} = 1(f_1)$$

$$x \frac{\partial f_2}{\partial x} + y \frac{\partial f_2}{\partial y} = 0(f_2)$$

$$\therefore x \frac{\partial}{\partial x}(f_1 + f_2) + y \frac{\partial}{\partial y}(f_1 + f_2) = f_1 + 0$$

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = x f\left(\frac{y}{x}\right) \quad \left[\because f_1 = x f\left(\frac{y}{x}\right) \right]$$

2.5.2. Extension of Euler's Theorem

If $f(x, y)$ is a homogeneous function of x, y of degree n , then

$$x^2 \frac{\partial^2 f}{\partial x^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} + y^2 \frac{\partial^2 f}{\partial y^2} = n(n-1)f$$

$$\text{(i.e.) } \left(x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y} \right)^2 f = n(n-1)f$$

Proof:

$$\text{By Euler's theorem, } x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = nf \quad \dots (1)$$

Differentiating equation (1) partially with respect to x ,

$$\begin{aligned} x \frac{\partial^2 f}{\partial x^2} + \frac{\partial f}{\partial x}(1) + y \frac{\partial^2 f}{\partial x \partial y} + \frac{\partial f}{\partial y}(0) &= n \frac{\partial f}{\partial x} \\ x \frac{\partial^2 f}{\partial x^2} + y \frac{\partial^2 f}{\partial x \partial y} &= (n-1) \frac{\partial f}{\partial x} \end{aligned} \quad \dots (2)$$

Differentiating equation (1) partially with respect to y ,

$$\begin{aligned} x \frac{\partial^2 f}{\partial y \partial x} + \frac{\partial f}{\partial x}(0) + y \frac{\partial^2 f}{\partial y^2} + \frac{\partial f}{\partial y}(1) &= n \frac{\partial f}{\partial y} \\ y \frac{\partial^2 f}{\partial y^2} + x \frac{\partial^2 f}{\partial y \partial x} &= (n-1) \frac{\partial f}{\partial y} \end{aligned} \quad \dots (3)$$

Multiplying equation (2) by x and equation (3) by y and adding,

$$x^2 \frac{\partial^2 f}{\partial x^2} + xy \frac{\partial^2 f}{\partial x \partial y} + y^2 \frac{\partial^2 f}{\partial y^2} + xy \frac{\partial^2 f}{\partial y \partial x} = (n-1) \left[x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} \right]$$

Hence proved.

Points to Remember:

1. Extension of Euler's Theorem: If $f(x, y)$ is a homogeneous function of degree

n in x and y , then $\left(x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y}\right)^2 f = n(n-1) f$

Example 1: If $u = \frac{x^2 y^2}{x+y}$, prove that $x \frac{\partial^2 u}{\partial x^2} + y \frac{\partial^2 u}{\partial x \partial y} = 2 \frac{\partial u}{\partial x}$.

Solution:

$$\text{Given that } u = f(x, y) = \frac{x^2 y^2}{x+y} \quad \left[\begin{array}{l} \because f(tx, ty) = t^n f(x, y) \\ \text{By Euler's theorem} \end{array} \right]$$

$$\therefore f(tx, ty) = \frac{(tx)^2 (ty)^2}{tx + ty} = \frac{t^4 (x^2 y^2)}{t(x+y)} = t^3 f(x, y)$$

$\therefore u$ is a homogeneous function of degree 3 in x, y .

$$\text{By Euler's theorem, } x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 3u \quad \dots (1)$$

Differentiating equation (1) partially with respect to x ,

$$\begin{aligned} x \frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x} (1) + y \frac{\partial^2 u}{\partial x \partial y} &= 3 \frac{\partial u}{\partial x} \\ x \frac{\partial^2 u}{\partial x^2} + y \frac{\partial^2 u}{\partial x \partial y} &= 2 \frac{\partial u}{\partial x}. \end{aligned}$$

Example 2: If $f(x, y) = \cos^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{y}{x}\right)$, prove that $x^2 \frac{\partial^2 f}{\partial x^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} +$

$$y^2 \frac{\partial^2 f}{\partial y^2} = 0.$$

Solution:

$$\text{Consider } f(tx, ty) = \cos^{-1}\left(\frac{tx}{ty}\right) + \tan^{-1}\left(\frac{ty}{tx}\right) = t^0 \cos^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{y}{x}\right) = t^0 f(x, y)$$

$\therefore f$ is homogeneous of degree zero.

By Extension of Euler's theorem,

$$x^2 \frac{\partial^2 f}{\partial x^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} + y^2 \frac{\partial^2 f}{\partial y^2} = n(n-1) f$$

$$\therefore x^2 \frac{\partial^2 f}{\partial x^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} + y^2 \frac{\partial^2 f}{\partial y^2} = 0.$$

Example 3: If $f = x^2 \tan^{-1} \frac{y}{x} - y^2 \sin^{-1} \frac{x}{y}$, prove that $x^2 \frac{\partial^2 f}{\partial x^2} + y^2 \frac{\partial^2 f}{\partial y^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} = 2f$.

Solution:

Given that $f(x, y) = x^2 \tan^{-1} \frac{y}{x} - y^2 \sin^{-1} \frac{x}{y}$

$$f(tx, ty) = t^2 x^2 \tan^{-1} \frac{ty}{tx} - t^2 y^2 \sin^{-1} \frac{tx}{ty}$$

$$= t^2 \left[x^2 \tan^{-1} \frac{y}{x} - y^2 \sin^{-1} \frac{x}{y} \right]$$

$$f(tx, ty) = t^2 f(x, y)$$

$\therefore f$ is homogeneous function of x, y of degree 2.

By Extension of Euler's theorem,

$$x^2 \frac{\partial^2 f}{\partial x^2} + y^2 \frac{\partial^2 f}{\partial y^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} = n(n-1) f = 2(2-1) f$$

$$x^2 \frac{\partial^2 f}{\partial x^2} + y^2 \frac{\partial^2 f}{\partial y^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} = 2f.$$

Example 4: If $z = xf\left(\frac{y}{x}\right) + g\left(\frac{y}{x}\right)$, prove that $x^2 \frac{\partial^2 z}{\partial x^2} + 2xy \frac{\partial^2 z}{\partial x \partial y} + y^2 \frac{\partial^2 z}{\partial y^2} = 0$.

Solution:

Let $z = f_1(x, y) + f_2(x, y)$ Where $f_1 = xf\left(\frac{y}{x}\right)$ and $f_2 = g\left(\frac{y}{x}\right)$

Here f_1 is a homogeneous function of degree 1 and f_2 is a homogeneous function of degree 0 in x, y .

By extension of Euler's theorem, $x^2 \frac{\partial^2 f}{\partial x^2} + y^2 \frac{\partial^2 f}{\partial y^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} = n(n-1)f$

$$\therefore x^2 \frac{\partial^2 f_1}{\partial x^2} + y^2 \frac{\partial^2 f_1}{\partial y^2} + 2xy \frac{\partial^2 f_1}{\partial x \partial y} = 1(1-1)f = 0 \quad \dots (1)$$

$$\text{Also, } x^2 \frac{\partial^2 f_1}{\partial x^2} + y^2 \frac{\partial^2 f_1}{\partial y^2} + 2xy \frac{\partial^2 f_1}{\partial x \partial y} = (0)f = 0 \quad \dots (2)$$

Adding equations (1) and (2) we get,

$$x^2 \frac{\partial^2}{\partial x^2} (f_1 + f_2) + 2xy \frac{\partial^2}{\partial x \partial y} (f_1 + f_2) + y^2 \frac{\partial^2}{\partial y^2} (f_1 + f_2) = 0$$

$$\text{Hence } x^2 \frac{\partial^2 z}{\partial x^2} + 2xy \frac{\partial^2 z}{\partial x \partial y} + y^2 \frac{\partial^2 z}{\partial y^2} = 0.$$

Example 5: If $u = \sin^{-1} \left(\frac{x^{\frac{1}{3}} + y^{\frac{1}{3}}}{x^{\frac{1}{2}} + y^{\frac{1}{2}}} \right)^{\frac{1}{2}}$, **Prove that** $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2}$

$$= \frac{\tan u}{144} (13 + \tan^2 u).$$

Solution:

Since u is not a homogeneous function.

$$\sin u = f(x, y) = \left(\frac{x^{\frac{1}{3}} + y^{\frac{1}{3}}}{x^{\frac{1}{2}} + y^{\frac{1}{2}}} \right)^{\frac{1}{2}}$$

$$f(tx, ty) = \left[\frac{t^{\frac{1}{3}} \left(x^{\frac{1}{3}} + y^{\frac{1}{3}} \right)}{t^{\frac{1}{2}} \left(x^{\frac{1}{2}} + y^{\frac{1}{2}} \right)} \right]^{\frac{1}{2}} = \left[t^{\frac{1}{6}} \frac{\left(x^{\frac{1}{3}} + y^{\frac{1}{3}} \right)}{\left(x^{\frac{1}{2}} + y^{\frac{1}{2}} \right)} \right]^{\frac{1}{2}} = t^{\frac{1}{12}} f(x, y)$$

$\therefore \sin u$ since is a homogeneous function of degree $-\frac{1}{12}$ in x and y .

By Euler's theorem, we have $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = nf$.

$$\begin{aligned} x \frac{\partial}{\partial x} (\sin u) + y \frac{\partial}{\partial y} (\sin u) &= \frac{-1}{12} \sin u \\ x \cos u \frac{\partial u}{\partial x} + y \cos u \frac{\partial u}{\partial y} &= \frac{-1}{12} \sin u \end{aligned}$$

Dividing by $\cos u$, $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{-1}{12} \tan u$... (1)

Differentiating equation (1) partially with respect to 'x'

$$\begin{aligned} x \frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x} (1) + y \frac{\partial^2 u}{\partial x \partial y} &= -\frac{1}{12} \sec^2 u \frac{\partial u}{\partial x} \\ x \frac{\partial^2 u}{\partial x^2} + y \frac{\partial^2 u}{\partial x \partial y} &= \frac{\partial u}{\partial x} \left[-\frac{1}{12} \sec^2 u - 1 \right] \end{aligned} \quad \dots (2)$$

Differentiating equation (1) partially with respect to 'y',

$$\begin{aligned} x \frac{\partial^2 u}{\partial x \partial y} + y \frac{\partial^2 u}{\partial y^2} + \frac{\partial u}{\partial y} &= -\frac{1}{12} \sec^2 u \frac{\partial u}{\partial y} \\ x \frac{\partial^2 u}{\partial x \partial y} + y \frac{\partial^2 u}{\partial y^2} &= \frac{\partial u}{\partial y} \left[-\frac{1}{12} \sec^2 u - 1 \right] \end{aligned} \quad \dots (3)$$

Multiplying equation (2) by x and equation (3) by y and adding we get,

$$\begin{aligned} x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} &= \left[-\frac{1}{12} \sec^2 u - 1 \right] \left[x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \right] \\ &= \left[-\frac{1}{12} \sec^2 u - 1 \right] \left[\frac{-1}{12} \tan u \right] = \frac{1}{144} \tan u \sec^2 u + \frac{1}{12} \tan u \\ &= \tan u \left[\frac{\tan^2 u + 1 + 12}{144} \right] \end{aligned}$$

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \frac{\tan u}{144} (13 + \tan^2 u)$$

Exercises

1. If $u = \sin^{-1} \left(\frac{x^2 + y^2}{x + y} \right)$, Prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \tan u$.
2. If $u = \log \left(\frac{x^4 + y^4}{x + y} \right)$, Prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 3$.
3. If $u = \frac{y}{z} + \frac{z}{x} + \frac{x}{y}$, Prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$
4. If $z = \log(x^2 + xy + y^2)$, Prove that $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 2$.
5. $z = \tan^{-1} \frac{y}{x}$, Prove that $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$
6. $f(x, y) = \log(x^2 + y^2) + \tan^{-1} \left(\frac{y}{x} \right)$. Prove that $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0$.
7. If $u = \frac{x}{r^3}$ and $r^2 = x^2 + y^2 + z^2$, Prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$.
8. If $u = \sin^{-1} \left(\frac{x + 2y + 3z}{\sqrt{x^8 + y^8 + z^8}} \right)$. Prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} + 3 \tan u = 0$
9. If $u = \tan^{-1} \left(\frac{x^3 + y^3}{x - y} \right)$, Prove that $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \sin 4u - \sin 2u$.
10. If $u = \frac{x + y}{x^2 + y^2}$, applying Euler's theorem, prove that the order of u is -1.

2.6 TOTAL DERIVATIVE

Let $z = f(x, y)$ be a function of two independent variables x and y . If the increments for x and y are considered as Δx and Δy , there will be an increment Δz for z .

$$\therefore z + \Delta z = f(x + \Delta x, y + \Delta y)$$

$$\Delta z = f(x + \Delta x, y + \Delta y) - f(x, y)$$

$$= \{f(x + \Delta x, y + \Delta y) - f(x, y + \Delta y)\} + \{f(x, y + \Delta y) - f(x, y)\}$$

Another manipulation is of the form,

$$\Delta z = \frac{f(x + \Delta x, y + \Delta y) - f(x + \Delta x, y)}{\Delta y} \Delta y + \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x} \Delta x$$

As $\Delta y \rightarrow 0$ and $\Delta x \rightarrow 0$, then we have

$$\Delta z = \left\{ \frac{\partial}{\partial y} f(x, y) \right\} \Delta y + \varepsilon \cdot \Delta y + \left\{ \frac{\partial}{\partial x} f(x, y) \right\} \Delta x + n \cdot \Delta x \quad [\because z = f(x, y)]$$

$$\Delta z = \frac{\partial z}{\partial y} \Delta y + \frac{\partial z}{\partial x} \Delta x + \varepsilon \cdot \Delta y + \eta \cdot \Delta x \quad \dots (4)$$

As Δx and Δy tends to zero independently, the first two terms on the R.H.S of (4) gives us an approximation to the increment Δz . This approximation is called the total differential of z and is denoted by dz .

$$\therefore dz = \frac{\partial z}{\partial x} \Delta x + \frac{\partial z}{\partial y} \Delta y.$$

In case of one independent variable, $dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$.

Points to Remember:

1. In case of three independent variables x, y and z , the total derivative is

$$\text{given by } dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy + \frac{\partial z}{\partial z} dz.$$

2. If z is a function of x & y and x and y themselves are functions of t , then

$$\text{we get } \frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}$$

Example 1: Find $\frac{du}{dx}$ if $u = \sin(x^2 + y^2)$ where $a^2x^2 + b^2y^2 = c^2$

Solution:

Given that $u = \sin(x^2 + y^2)$

Differentiating with respect to x , we get

$$\begin{aligned}\frac{du}{dx} &= \frac{\partial u}{\partial x} \cdot \frac{dx}{dx} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dx} \quad \because du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy. \\ &= 2x \cos(x^2 + y^2) + 2y \cos(x^2 + y^2) \frac{dy}{dx} \quad \dots(1)\end{aligned}$$

Given that $a^2x^2 + b^2y^2 = c^2$.

Differentiating the above equation with respect to ' x ',

$$2a^2x + 2b^2y \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{a^2x}{b^2y} \quad \dots(2)$$

Using equation (2) in (1), we get

$$\begin{aligned}\frac{du}{dx} &= 2x \cos(x^2 + y^2) + 2y \cos(x^2 + y^2) \left(-\frac{a^2x}{b^2y} \right) \\ \frac{du}{dx} &= 2x \cos(x^2 + y^2) + 2y \cos(x^2 + y^2) \left(-\frac{a^2x}{b^2y} \right) \\ &= 2x \cos(x^2 + y^2) - 2 \frac{a^2}{b^2} x \cos(x^2 + y^2) \\ \therefore \frac{du}{dx} &= 2x \cos(x^2 + y^2) \left[\frac{b^2 - a^2}{b^2} \right]\end{aligned}$$

Example 2: If $u = xy + yz + zx$ where $x = e^t, y = e^{-t}$ and $z = \frac{1}{t}$, find $\frac{du}{dt}$.

Solution:

Given that $u = xy + yz + zx$

$$\begin{aligned}
 \frac{du}{dt} &= \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial u}{\partial z} \cdot \frac{dz}{dt} \\
 &= (y+z) \cdot e^t + (z+x)(-e^{-t}) + (x+y) \left(-\frac{1}{t^2} \right) \\
 &= \left(e^{-t} + \frac{1}{t} \right) e^t + \left(\frac{1}{t} + e^t \right) (-e^{-t}) + (e^t + e^{-t}) \left(-\frac{1}{t^2} \right) \\
 &= 1 + \frac{e^t}{t} - \frac{e^{-t}}{t} - 1 - \frac{e^t}{t^2} - \frac{e^{-t}}{t^2} = \frac{1}{t} (2 \sinh t) - \frac{1}{t^2} (2 \cosh t) \\
 \frac{du}{dt} &= \frac{2}{t^2} [t \sinh t - \cosh t]
 \end{aligned}$$

Example 3: Find $\frac{du}{dt}$ when $u = xy^2 + x^2y, x = at^2, y = 2at$. Also verify by direct substitution.

Solution:

$$\begin{aligned}
 \text{We know that } \frac{du}{dt} &= \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} \\
 &= (y^2 + 2xy)(2at) + (2xy + x^2)(2a) \\
 &= [4a^2t^2 + 2at^2(2at)](2at) + (2at^2 \cdot 2at + a^2t^4)(2a) \\
 &\quad \left[\because x = at^2, y = 2at \right] \\
 &= 8a^3t^3 + 8a^3t^4 + 8a^3t^3 + 2a^3t^4 = 16a^3t^3 + 10a^3t^4 \\
 \therefore \frac{du}{dt} &= 2a^3t^3(8+5t)
 \end{aligned}$$

Verification: $u = at^2(4a^2t^2) + a^2t^4(2at)$

$$= 4a^3t^4 + 2a^3t^5$$

$$\frac{du}{dt} = 16a^3t^3 + 10a^3t^4 = 2a^3t^3(8+5t)$$

Example 4: If $u = \sin^{-1}(x-y)$, $x = 3t$, $y = 4t^3$, **Prove that** $\frac{du}{dt} = \frac{3}{\sqrt{1-t^2}}$.

Solution:

$$\begin{aligned} \text{We know that } \frac{du}{dt} &= \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} \\ &= \frac{1}{\sqrt{1-(x-y)^2}}(3) + \frac{1}{\sqrt{1-(x-y)^2}}(-1)(12t^2) \\ &= \frac{3-12t^2}{\sqrt{1-(x-y)^2}} = \frac{3(1-4t^2)}{\sqrt{1-(3t-4t^3)^2}} \\ &= \frac{3(1-4t^2)}{\sqrt{1-9t^2+24t^4-16t^6}} = \frac{3(1-4t^2)}{\sqrt{(1-t^2)(1-8t^2+16t^4)}} \\ &= \frac{3(1-4t^2)}{\sqrt{(1-t^2)(1-4t^2)^2}} = \frac{3}{\sqrt{1-t^2}} \end{aligned}$$

Example 5: If $u = \sin\left(\frac{x}{y}\right)$, $x = e^t$, $y = t^2$, **Find** $\frac{du}{dt}$ **as a function of** t .

Solution:

$$\text{We know that } \frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} \quad \dots (1)$$

$$u = \sin\left(\frac{x}{y}\right)$$

$$\frac{\partial u}{\partial x} = \cos\left(\frac{x}{y}\right) \cdot \left(\frac{1}{y}\right) ; \frac{\partial u}{\partial y} = \cos\left(\frac{x}{y}\right) \cdot \left(\frac{-x}{y^2}\right)$$

$$\text{Also } x = e^t, y = t^2$$

$$\frac{dx}{dt} = e^t, \quad \frac{dy}{dt} = 2t$$

Hence equation (1) becomes,
$$\frac{du}{dt} = \frac{1}{y} \cos\left(\frac{x}{y}\right) \cdot (e^t) + \cos\left(\frac{x}{y}\right) \left(\frac{-x}{y^2}\right) \cdot (2t)$$

$$= \frac{e^t}{t^2} \cos\left(\frac{e^t}{t^2}\right) - \frac{2e^t t}{t^4} \cos\left(\frac{e^t}{t^2}\right) = \frac{e^t(t-2)}{t^3} \cdot \cos\left(\frac{e^t}{t^2}\right)$$

Example 6: If $z = xy(x+y)$, find d^2z .

Solution:

If $z = f(x, y)$, then $dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$

$$dz = \left(dx \frac{\partial}{\partial x} + dy \frac{\partial}{\partial y} \right) z = d(z)$$

$$d^2z = d(dz) = \left(dx \frac{\partial}{\partial x} + dy \frac{\partial}{\partial y} \right) \left(dx \frac{\partial}{\partial x} + dy \frac{\partial}{\partial y} \right) z$$

$$= \left(dx \frac{\partial}{\partial x} + dy \frac{\partial}{\partial y} \right)^2 z$$

$$= \left[(dx)^2 \frac{\partial^2}{\partial x^2} + (dy)^2 \frac{\partial^2}{\partial y^2} + 2dx dy \frac{\partial^2}{\partial x \partial y} \right] z$$

$$d^2z = \frac{\partial^2 z}{\partial x^2} (dx)^2 + \frac{\partial^2 z}{\partial y^2} (dy)^2 + 2 \frac{\partial^2 z}{\partial x \partial y} dx dy \quad \dots (1)$$

Since $z = xy(x+y) = x^2y + xy^2$

$$\frac{\partial z}{\partial x} = 2xy + y^2, \quad \frac{\partial z}{\partial y} = x^2 + 2xy$$

$$\frac{\partial^2 z}{\partial x^2} = 2y, \quad \frac{\partial^2 z}{\partial y^2} = 2x, \quad \frac{\partial^2 z}{\partial x \partial y} = 2x + 2y$$

$\therefore$ Equation (1) becomes, $d^2z = 2y(dx)^2 + 2x(dy)^2 + 4(x+y)dx dy$.

Example 7: If $x^2 + y^2 + z^2 - 2xyz = 1$, Prove that $\frac{dx}{\sqrt{1-x^2}} + \frac{dy}{\sqrt{1-y^2}} + \frac{dz}{\sqrt{1-z^2}} = 0$.

Solution:

Given that $x^2 + y^2 + z^2 - 2xyz = 1$... (1)

Taking differentials, we get

$$2x dx + 2y dy + 2z dz - 2yz dx - 2xz dy - 2xy dz = 0$$

$$(x - yz) dx + (y - xz) dy + (z - xy) dz = 0 \quad \dots (2)$$

Now, $(x - yz)^2 = x^2 - 2xyz + y^2 z^2 \quad (\because x^2 - 2xyz = 1 - y^2 - z^2)$

$$= 1 - y^2 - z^2 + y^2 z^2 = (1 - y^2) - z^2 (1 - y^2)$$

$$= (1 - y^2)(1 - z^2)$$

$$x - yz = \sqrt{(1 - y^2)(1 - z^2)}$$

Similarly, $y - zx = \sqrt{(1 - z^2)(1 - x^2)}$

$$(z - xy) = \sqrt{(1 - x^2)(1 - y^2)}$$

Using these values in equation (2), we get

$$\sqrt{(1 - y^2)(1 - z^2)} dx + \sqrt{(1 - z^2)(1 - x^2)} dy + \sqrt{(1 - x^2)(1 - y^2)} dz = 0$$

Dividing by $\sqrt{(1 - x^2)(1 - y^2)(1 - z^2)}$

$$\frac{dx}{\sqrt{1-x^2}} + \frac{dy}{\sqrt{1-y^2}} + \frac{dz}{\sqrt{1-z^2}} = 0$$

Example 8: If the sides and angles of a triangle ABC vary in such a way that its circumradius remains constant, prove that $\frac{da}{\cos A} + \frac{db}{\cos B} + \frac{dc}{\cos C} = 0$.

Solution:

The circum-Radius R of ΔABC is given by $R = \frac{a}{2 \sin A} = \frac{b}{2 \sin B} = \frac{c}{2 \sin C}$

Now $a = 2R \sin A$ (R is constant).

Taking differentials, $da = 2R \cos A dA$

$$\therefore \frac{da}{\cos A} = 2R dA$$

Similarly, $\frac{db}{\cos B} = 2R dB$ and $\frac{dc}{\cos C} = 2R dC$.

Adding we get, $\frac{da}{\cos A} + \frac{db}{\cos B} + \frac{dc}{\cos C} = 2R(dA + dB + dC)$

We know that $A + B + C = \pi$

Taking differentials, $dA + dB + dC = 0$.

$$\therefore \frac{da}{\cos A} + \frac{db}{\cos B} + \frac{dc}{\cos C} = 0$$

Example 9: If $u = xyz$, find d^2u .

Solution: If $u = f(x, y, z) = xyz$, then

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy + \frac{\partial u}{\partial z} dz.$$

$$d(u) = \left(dx \frac{\partial}{\partial x} + dy \frac{\partial}{\partial y} + dz \frac{\partial}{\partial z} \right) u.$$

$$d^2u = d(du)$$

$$= \left(dx \frac{\partial}{\partial x} + dy \frac{\partial}{\partial y} + dz \frac{\partial}{\partial z} \right) \left(dx \frac{\partial}{\partial x} + dy \frac{\partial}{\partial y} + dz \frac{\partial}{\partial z} \right) u$$

$$= \left(dx \frac{\partial}{\partial x} + dy \frac{\partial}{\partial y} + dz \frac{\partial}{\partial z} \right)^2 u$$

$$= (dx)^2 \frac{\partial^2 u}{\partial x^2} + (dy)^2 \frac{\partial^2 u}{\partial y^2} + (dz)^2 \frac{\partial^2 u}{\partial z^2} + 2dx dy \frac{\partial^2 u}{\partial x \partial y} + 2dy dz \frac{\partial^2 u}{\partial y \partial z} + 2dx dz \frac{\partial^2 u}{\partial x \partial z} \dots (1)$$

Since $u = xyz$, $\frac{\partial u}{\partial x} = yz$; $\frac{\partial u}{\partial y} = zx$; $\frac{\partial u}{\partial z} = xy$

$$\frac{\partial^2 u}{\partial x^2} = 0 ; \frac{\partial^2 u}{\partial y^2} = 0 ; \frac{\partial^2 u}{\partial z^2} = 0$$

$$\frac{\partial^2 u}{\partial x \partial y} = z ; \frac{\partial^2 u}{\partial y \partial z} = x ; \frac{\partial^2 u}{\partial x \partial z} = y$$

Substituting in equation (1) we get $d^2u = 0 + 0 + 0 + 2zdx dy + 2xdy dz + 2ydx dz$
 $\therefore d^2u = 2x dy dz + 2y dz dx + 2z dx dy$.

Exercises

1. If $u = x^3 y^4$ where $x = t^3, y = t^2$, find $\frac{du}{dt}$.
2. If $u = x^2 y^3$ where $x = \log t$ and $y = e^t$, find $\frac{du}{dt}$.
3. If $u = \tan(3x - y) + 6^{y+z}$, find du .
4. If $z = xy^2 + yx^2$, $x = at^2$, $y = 2at$. Find $\frac{dz}{dt}$.
5. If $u = x^2 + y^2 + z^2$, $x = e^t \cos t, y = e^t \sin t, z = e^t$ Find $\frac{du}{dt}$.
6. If $u = e^x \sin yz$, $x = t^2, y = t - 1, z = \frac{1}{t}$. Find $\frac{du}{dt}$.
7. In a plane triangle, if the sides a, b, c are constant. Prove that the variations of its angles are given by the relations $\frac{dA}{\sqrt{a^2 - b^2 \sin^2 A}} = \frac{dB}{\sqrt{b^2 - a^2 \sin^2 B}} = \frac{-dC}{c}$.

Answers

1. $17t^{16}$
2. $\frac{2C^{3t} \log t}{t} + 3e^{3t} (\log t)^2$
3. $du = 3 \sec^2(3x - y) dx + [6^{y+z} (\log 6) - \sec^2(3x - y)] dy + 6^{y+z} (\log 6) dz$
4. $a^3 (16t^3 + 10t^4)$
5. $4e^{2t}$
6. $e^{t^2} \left[2t \sin \left(\frac{t-1}{t} \right) + \frac{1}{t^2} \cos \left(\frac{t-1}{t} \right) \right]$

2.7 DIFFERENTIATION OF IMPLICIT FUNCTIONS

Implicit Function: Let f be a function of two variables x and y on solving $f(x, y) = 0$, this defines y as an implicit function of x .

Derivative of Implicit Function: Let $f(x, y) = 0$ be defined for y as an implicit function of x .

$$\therefore \frac{df}{dx} = \frac{\partial f}{\partial x} \frac{dx}{dx} + \frac{\partial f}{\partial y} \frac{dy}{dx}$$

$$\text{However, } \frac{df}{dx} = 0 \Rightarrow \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \frac{dy}{dx} = 0.$$

$$\frac{dy}{dx} = \frac{-\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}} = \frac{-f_x}{f_y} \text{ if } y \neq 0.$$

$$\text{Similarly, } \frac{d^2y}{dx^2} = -\frac{\left[f_{xx}(f_y)^2 - 2f_{xy}f_xf_y + f_{yy}(f_x)^2 \right]}{(f_y)^3}$$

$$\frac{d^2y}{dx^2} = -\frac{(p^2t - 2pqs + q^2r)}{q^3} \quad \text{where } p = f_x, q = f_y, r = f_{xx}, s = f_{xy}, t = f_{yy}$$

Example 1: Find $\frac{dy}{dx}$ if $x^y + y^x = a^k$.

Solution:

$$\text{We know that } \frac{d(a^x)}{dx} = a^x \log a, \quad \frac{d(x^a)}{dx} = ax^{a-1}.$$

$$f(x, y) = x^y + y^x - a^k = 0$$

$$f_x = yx^{y-1} + y^x \log y$$

$$f_y = x^y \log x + xy^{x-1}$$

$$\therefore \frac{dy}{dx} = -\frac{f_x}{f_y} = \frac{yx^{y-1} + y^x \log y}{x^y \log x + xy^{x-1}}.$$

Example 2: Find $\frac{dy}{dx}$ if $x^3 + y^3 = 3ax^2y$.

Solution:

Let $f(x, y) = x^3 + y^3 - 3ax^2y$

$$\frac{\partial f}{\partial x} = f_x = 3x^2 - 6axy ; \quad \frac{\partial f}{\partial y} = f_y = 3y^2 - 3ax^2$$

$$\frac{dy}{dx} = -\frac{f_x}{f_y} = \frac{-(3x^2 - 6axy)}{3y^2 - 3ax^2} = \frac{3x(2ay - x)}{3(y^2 - ax^2)}$$

$$\therefore \frac{dy}{dx} = \frac{x(2ay - x)}{(y^2 - ax^2)} .$$

Example 3: If $y^3 - 3ax^2 + x^3 = 0$. Prove that $\frac{d^2y}{dx^2} + \frac{2a^2x^2}{y^5} = 0$.

Solution:

Let $f(x, y) = y^3 - 3ax^2 + x^3$

$$f_x = -6ax + 3x^2 ; \quad f_y = 3y^2$$

$$f_{xx} = -6a + 6x ; f_{xy} = 0 ; \quad f_{yy} = 6y$$

$$\frac{d^2y}{dx^2} = \frac{f_{xx}(f_y)^2 - 2f_{xy}f_xf_y + f_{yy}(f_x)^2}{(f_y)^3}$$

$$= \frac{\left[(6x - 6a)(9y^4) - 0 + 6y(3x^2 - 6ax)^2 \right]}{27y^6}$$

$$= \frac{-54y \left[(x - a)y^3 + (x^2 - 2ax)^2 \right]}{27y^6}$$

$$\begin{aligned}
&= \frac{-2 \left[(x-a)(3ax^2 - x^3) + (x^2 - 2ax)^2 \right]}{y^5} \quad (\because y^3 = 3ax^2 - x^3) \\
&= -2 \frac{a^2 x^2}{y^5} \\
y'' + 2 \frac{a^2 x^2}{y^5} &= 0
\end{aligned}$$

Example 4: If $ax^2 + 2hxy + by^2 = 1$, Prove that $\frac{d^2y}{dx^2} = \frac{h^2 - ab}{(hx + by)^3}$.

Solution:

Let $f(x, y) = ax^2 + 2hxy + by^2 - 1$

$$f_x = 2ax + 2hy = 2(ax + hy), \quad f_y = 2hx + 2by = 2(hx + by)$$

$$f_{xx} = 2a; \quad f_{xy} = 2h; \quad f_{yy} = 2b;$$

$$\frac{dy}{dx} = -\frac{f_x}{f_y} = -\frac{2(ax + hy)}{2(hx + by)} = -\frac{(ax + hy)}{(hx + by)}$$

$$\begin{aligned}
\frac{d^2y}{dx^2} &= \frac{-\left[f_{xx}(f_y)^2 - 2f_{xy}f_xf_y + f_{yy}(f_x)^2 \right]}{(f_y)^3} \\
&= \frac{-\left\{ 2a(2(hx + by))^2 - 2(2h)2(ax + hy)2(hx + by) + 2b(2(ax + hy))^2 \right\}}{(2(hx + by))^3}
\end{aligned}$$

$$\begin{aligned}
 &= \frac{-\left\{8a(h^2x^2 + b^2y^2 + 2hbx y) - 16h(ahx^2 + h^2xy + abxy + hby^2)\right\} + 8b(a^2x^2 + h^2y^2 + 2ahxy)}{8(hx + by)^3} \\
 &= \frac{-8\left\{ah^2x^2 + ab^2y^2 + 2abhx y - 2ah^2x^2 - 2h^3xy - 2abhx y - 2h^2by^2\right\} + a^2bx^2 + bh^2y^2 + 2abhx y}{8(hx + by)^3} \\
 &= \frac{\{a(h^2 - ab)x^2 + 2h(h^2 - ab)xy + b(h^2 - ab)y^2\}}{(hx + by)^3} \\
 &= \frac{\{(h^2 - ab)(ax^2 + 2hxy + by^2)\}}{(hx + by)^3} \quad \left[\because ax^2 + 2hxy + by^2 = 1\right]
 \end{aligned}$$

$$\frac{d^2y}{dx^2} = \frac{h^2 - ab}{(hx + by)^3}$$

Example 5: If $(\cos x)^y = (\sin y)^x$, find $\frac{dy}{dx}$.

Solution:

Given that $(\cos x)^y = (\sin y)^x$

Taking logarithm on both sides, $y \log(\cos x) = x \log(\sin y)$

Let $f(x, y) = y \log(\cos x) - x \log(\sin y) = 0$

$$f_x = -y \tan x - \log \sin y, \quad f_y = \log \cos x - x \cot y$$

$$\therefore \frac{dy}{dx} = \frac{-f_x}{f_y} = \frac{y \tan x + \log \sin y}{\log \cos x - x \cot y}$$

Example 6: If $u = \cos(x^2 + y^2)$ where $a^2x^2 + b^2y^2 = c^2$. Find $\frac{du}{dx}$.

Solution:

Given that $u = \cos(x^2 + y^2)$

$$\frac{\partial u}{\partial x} = -\sin(x^2 + y^2) \cdot (2x)$$

$$\frac{\partial u}{\partial y} = -\sin(x^2 + y^2) \cdot (2y)$$

Let $f = a^2x^2 + b^2y^2 - c^2$.

$$\frac{dy}{dx} = -\frac{f_x}{f_y} = -\frac{2a^2x}{2b^2y} = -\frac{a^2x}{b^2y}$$

$$\frac{du}{dx} = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dx}$$

$$\frac{du}{dx} = -\sin(x^2 + y^2) \cdot (2x) + (-2y \sin(x^2 + y^2)) \cdot \left(-\frac{a^2x}{b^2y}\right) = 2x \sin(x^2 + y^2) \left[\frac{a^2}{b^2} - 1\right]$$

$$\frac{du}{dx} = 2x \left[\frac{a^2}{b^2} - 1\right] \sin(x^2 + y^2).$$

Example 7: Find $\frac{du}{dx}$ if $u = \tan^{-1}\left(\frac{x}{y}\right)$ where $x^2 + y^2 = a^2$.

Solution:

Given that $u = \tan^{-1}\left(\frac{x}{y}\right)$ and $f = x^2 + y^2 - a^2$

$$\frac{dy}{dx} = -\frac{f_x}{f_y} = -\frac{2x}{2y} = -\frac{x}{y} \quad \dots (1)$$

$$\text{We know that } du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy \quad \dots (2)$$

$$\frac{\partial u}{\partial x} = \frac{1}{1 + \frac{x^2}{y^2}} \cdot \left(\frac{1}{y}\right) = \frac{y}{x^2 + y^2} \quad \dots (3)$$

$$\frac{\partial u}{\partial y} = \frac{1}{1 + \frac{x^2}{y^2}} \cdot \left(\frac{-x}{y^2} \right) = \frac{-x}{x^2 + y^2} \quad \dots (4)$$

Substituting equations (1), (3), (4) in (2), we get

$$\begin{aligned} \frac{du}{dx} &= \frac{y}{x^2 + y^2} - \frac{x}{(x^2 + y^2)} \left(-\frac{x}{y} \right) = \frac{y}{x^2 + y^2} + \frac{x^2}{y(x^2 + y^2)} \\ \therefore \frac{du}{dx} &= \frac{y^2 + x^2}{y(x^2 + y^2)} = \frac{1}{y} \end{aligned}$$

Example 8: If $x^3 + y^3 - 3axy = 0$, Find $\frac{d^2y}{dx^2}$.

Solution:

Given that $f(x, y) = x^3 + y^3 - 3axy$

$$f_x = 3x^2 - 3ay; \quad f_y = 3y^2 - 3ax; \quad f_{xy} = -3a$$

$$f_{xx} = 6x; \quad f_{yy} = 6y$$

$$\frac{dy}{dx} = -\frac{f_x}{f_y} = -\left(\frac{3x^2 - 3ay}{3y^2 - 3ax} \right) = \left(\frac{x^2 - ay}{ax - y^2} \right)$$

$$\frac{d^2y}{dx^2} = -\frac{f_{xx}f_y^2 - 2f_xf_yf_{xy} + f_{yy}f_x^2}{f_y^3}$$

$$\frac{d^2y}{dx^2} = -\frac{6x(3y^2 - 3ax)^2 - 2(3x^2 - 3ay)(3y^2 - 3ax)(-3a) + 6y(3x^2 - 3ay)^2}{(3y^2 - 3ax)^3}$$

$$= -\frac{54}{27(y^2 - ax)^3} \left[x(y^2 - ax)^2 + a(x^2 - ay)(y^2 - ax) + y(x^2 - ay)^2 \right]$$

$$= \frac{2}{(ax - y^2)^3} \left[xy^4 - 2ax^2y^2 + a^2x^3 + ax^2y^2 - a^2x^3 - a^2y^3 + a^3xy + yx^4 - 2ax^2y^2 + a^2y^3 \right]$$

$$\begin{aligned}
 &= \frac{2}{(ax - y^2)^3} [xy^4 + x^4y - 3ax^2y^2 + a^3xy] \\
 &= \frac{2}{(ax - y^2)^3} [xy(x^3 + y^3 - 3axy) + a^3xy] \quad \left[\because x^3 + y^3 - 3axy = 0 \right] \\
 \frac{d^2y}{dx^2} &= \frac{2a^3xy}{(ax - y^2)^3}
 \end{aligned}$$

Exercises

- Find $\frac{dy}{dx}$ if $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$.
- Find $\frac{dy}{dx}$ if $x^y = y^x$.
- Find $\frac{dy}{dx}$ if $x^3 + 3x^2y + 6xy^2 + y^3 = 1$.
- Find $\frac{du}{dx}$ if $u = x \log xy$ where $x^3 + y^3 + 3xy = 1$.
- Find $\frac{du}{dx}$ if $u = x^2y$ where $x^2 + xy + y^2 = 1$.
- Find $\frac{d^2y}{dx^2}$ if $ay^2 + 2by + c = x^2$
- Find $\frac{d^2y}{dx^2}$ if $x^4 + y^4 = 4a^2xy$.

Answers

- $\frac{-(ax + hy + g)}{(hx + by + f)}$
- $\frac{y(y - x \log y)}{x(x - y \log x)}$
- $\frac{-(x^2 + 2xy + 2y^2)}{(x^2 + 4xy + y^2)}$
- $1 + \log(xy) - \frac{x(x^2 + y)}{y(x + y^2)}$

$$5. \frac{x(xy + 4y^2 + 2x^2)}{(x + 2y)}$$

$$7. \frac{2a^2xy(3a^4 + x^2y^2)}{(a^2x - y^3)^3}$$

$$6. \frac{b^2 - ac}{(ay + b)^3}$$

2.8 CHANGE OF VARIABLES

If $u = f(x, y)$... (1) where $x = \phi(s, t)$ and $y = \psi(s, t)$... (2).

It is often necessary to change expressions involving $u, x, y, \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}$ etc. to expressions

involving $u, s, t, \frac{\partial u}{\partial s}, \frac{\partial u}{\partial t}$ etc.

If t is regarded as a constant, then x, y, u will be functions of s alone.

$$\text{Hence we have } \frac{\partial u}{\partial s} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial s} \quad \dots(3)$$

Where the ordinary derivatives have been replaced by the partial derivatives because x, y are functions of two variables s and t . Similarly, regarding s as constant,

$$\text{We have } \frac{\partial u}{\partial t} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial t} \quad \dots(4)$$

On solving equations (3) and (4), we get $\frac{\partial u}{\partial x}$ and $\frac{\partial u}{\partial y}$ in terms of $\frac{\partial u}{\partial s}, \frac{\partial u}{\partial t}, u, s, t$.

Also equation (2) can be written as $s = \xi(x, y)$ and $t = \eta(x, y)$.

$$\text{It is easier to use } \frac{\partial u}{\partial x} = \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial x} + \frac{\partial u}{\partial t} \cdot \frac{\partial t}{\partial x}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial y} + \frac{\partial u}{\partial t} \cdot \frac{\partial t}{\partial y}$$

The higher derivatives of u can be found by repeated application of (3) and (4)

Example 1: If $x = u^2 - v^2, y = 2uv$. Find $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}$.

Solution:

$$\text{Given that } x = u^2 - v^2 \quad \dots(1) \qquad y = 2uv \quad \dots(2)$$

Differentiating equations (1) and (2) partially with respect to 'x' we get,

$$1 = 2u \frac{\partial u}{\partial x} - 2v \frac{\partial v}{\partial x} \quad ; \quad 0 = 2u \frac{\partial v}{\partial x} + 2v \frac{\partial u}{\partial x}$$

On solving we get,

$$\frac{\partial u}{\partial x} = \frac{u}{2(u^2 + v^2)} \quad \text{and} \quad \frac{\partial v}{\partial x} = \frac{-v}{2(u^2 + v^2)}$$

Differentiating equation s(1) and (2) partially with respect to 'y' we get,

$$0 = 2u \frac{\partial u}{\partial y} - 2v \frac{\partial v}{\partial y} \quad ; \quad 1 = 2v \frac{\partial u}{\partial y} + 2u \frac{\partial v}{\partial y}$$

On solving we get,

$$\frac{\partial u}{\partial y} = \frac{v}{2(u^2 + v^2)} \quad \text{and} \quad \frac{\partial v}{\partial y} = \frac{u}{2(u^2 + v^2)}$$

Example 2: If $u = f(r, s)$, $r = x + y$, $s = x - y$, prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 2 \frac{\partial u}{\partial r}$.

Solution:

Given that $u = f(r, s)$, $r = x + y$, $s = x - y$

$$\frac{\partial r}{\partial x} = 1; \frac{\partial r}{\partial y} = 1; \frac{\partial s}{\partial x} = 1; \frac{\partial s}{\partial y} = -1$$

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial x} + \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial x} \\ &= \frac{\partial u}{\partial r}(1) + \frac{\partial u}{\partial s}(1) = \frac{\partial u}{\partial r} + \frac{\partial u}{\partial s} \end{aligned} \quad \dots(1)$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial y} + \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial y}$$

$$= \frac{\partial u}{\partial r}(1) + \frac{\partial u}{\partial s}(-1) = \frac{\partial u}{\partial r} - \frac{\partial u}{\partial s} \quad \dots(2)$$

Adding equation (1) and (2) we get.

$$\begin{aligned} \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} &= \frac{\partial u}{\partial r} + \frac{\partial u}{\partial s} + \frac{\partial u}{\partial r} - \frac{\partial u}{\partial s} = 2 \frac{\partial u}{\partial r} \\ \therefore \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} &= 2 \frac{\partial u}{\partial r} \end{aligned}$$

Example 3: If $u = f(x, y)$ where $x = r \cos \theta, y = r \sin \theta$ Prove that

$$\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 = \left(\frac{\partial u}{\partial r} \right)^2 + \frac{1}{r^2} \left(\frac{\partial u}{\partial \theta} \right)^2.$$

Solution:

Given that $u = f(x, y)$

$$\begin{aligned} \frac{\partial u}{\partial r} &= \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial r} = \frac{\partial u}{\partial x}(\cos \theta) + \frac{\partial u}{\partial y}(\sin \theta) \\ \frac{\partial u}{\partial \theta} &= \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial \theta} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial \theta} = \frac{\partial u}{\partial x}(-r \sin \theta) + \frac{\partial u}{\partial y}(r \cos \theta) \\ &= r \left(-\frac{\partial u}{\partial x} \sin \theta + \frac{\partial u}{\partial y} \cos \theta \right) \end{aligned}$$

From equation (1) and (2),

$$\begin{aligned} \left(\frac{\partial u}{\partial r} \right)^2 + \frac{1}{r^2} \left(\frac{\partial u}{\partial \theta} \right)^2 &= \left(\cos \theta \frac{\partial u}{\partial x} + \sin \theta \frac{\partial u}{\partial y} \right)^2 + \left(-\sin \theta \frac{\partial u}{\partial x} + \cos \theta \frac{\partial u}{\partial y} \right)^2 \\ &= \left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 \\ \therefore \left(\frac{\partial u}{\partial r} \right)^2 + \frac{1}{r^2} \left(\frac{\partial u}{\partial \theta} \right)^2 &= \left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 \end{aligned}$$

Example 4: If $u = f(x - y, y - z, z - x)$, prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$.

Solution:

Let $x - y = p$, $y - z = q$, $z - x = r$.

$\therefore u = f(p, q, r)$ where p, q, r are functions of x, y, z .

$$\begin{aligned} \text{By chain rule, } \frac{\partial u}{\partial x} &= \frac{\partial u}{\partial p} \cdot \frac{\partial p}{\partial x} + \frac{\partial u}{\partial q} \cdot \frac{\partial q}{\partial x} + \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial x} \\ &= \frac{\partial u}{\partial p}(1) + \frac{\partial u}{\partial q}(0) + \frac{\partial u}{\partial r}(-1) \end{aligned}$$

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial u}{\partial p} - \frac{\partial u}{\partial r} \quad \dots(1)$$

$$\begin{aligned} \frac{\partial u}{\partial y} &= \frac{\partial u}{\partial p} \cdot \frac{\partial p}{\partial y} + \frac{\partial u}{\partial q} \cdot \frac{\partial q}{\partial y} + \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial y} \\ &= \frac{\partial u}{\partial p}(-1) + \frac{\partial u}{\partial q}(1) + \frac{\partial u}{\partial r}(0) \end{aligned}$$

$$\therefore \frac{\partial u}{\partial y} = -\frac{\partial u}{\partial p} + \frac{\partial u}{\partial q} \quad \dots(2)$$

$$\begin{aligned} \frac{\partial u}{\partial z} &= \frac{\partial u}{\partial p} \cdot \frac{\partial p}{\partial z} + \frac{\partial u}{\partial q} \cdot \frac{\partial q}{\partial z} + \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial z} \\ &= \frac{\partial u}{\partial p}(0) + \frac{\partial u}{\partial q}(-1) + \frac{\partial u}{\partial r}(1) \end{aligned}$$

$$\therefore \frac{\partial u}{\partial z} = -\frac{\partial u}{\partial q} + \frac{\partial u}{\partial r} \quad \dots(3)$$

Adding equations (1), (2), (3) we get,

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{\partial u}{\partial p} - \frac{\partial u}{\partial r} - \frac{\partial u}{\partial p} + \frac{\partial u}{\partial q} - \frac{\partial u}{\partial q} + \frac{\partial u}{\partial r} = 0.$$

Example 5: If $u = f(x, y)$ where $x = e^r \cos \theta$, $y = e^r \sin \theta$. Show that

$$x \frac{\partial u}{\partial \theta} + y \frac{\partial u}{\partial r} = e^{2r} \frac{\partial u}{\partial y}.$$

Solution:

Given that $u = f(x, y)$ where x and y are functions of r and θ .

$$\begin{aligned} \frac{\partial u}{\partial r} &= \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial r} \\ &= \frac{\partial u}{\partial x} \cdot (e^r \cos \theta) + \frac{\partial u}{\partial y} \cdot (e^r \sin \theta) \\ &= e^r \cos \theta \frac{\partial u}{\partial x} + e^r \sin \theta \frac{\partial u}{\partial y} = x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \end{aligned}$$

$$\begin{aligned} \frac{\partial u}{\partial \theta} &= \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial \theta} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial \theta} \\ &= \frac{\partial u}{\partial x} \cdot (-e^r \sin \theta) + \frac{\partial u}{\partial y} \cdot (e^r \cos \theta) \\ &= -e^r \sin \theta \frac{\partial u}{\partial x} + e^r \cos \theta \frac{\partial u}{\partial y} = -y \frac{\partial u}{\partial x} + x \frac{\partial u}{\partial y} \end{aligned}$$

$$\begin{aligned} \therefore x \frac{\partial u}{\partial \theta} + y \frac{\partial u}{\partial r} &= x \left(-y \frac{\partial u}{\partial x} + x \frac{\partial u}{\partial y} \right) + y \left(x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \right) \\ &= -xy \frac{\partial u}{\partial x} + x^2 \frac{\partial u}{\partial y} + xy \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} \end{aligned}$$

$$\therefore x \frac{\partial u}{\partial \theta} + y \frac{\partial u}{\partial r} = x^2 \frac{\partial u}{\partial y} + y^2 \frac{\partial u}{\partial y} = (x^2 + y^2) \frac{\partial u}{\partial y}$$

$$\text{Hence } x \frac{\partial u}{\partial \theta} + y \frac{\partial u}{\partial r} = e^{2r} \frac{\partial u}{\partial y} \quad \left[\because x^2 + y^2 = e^{2r} \right]$$

Example 6: If $u = f(r, s, t)$ and $r = \frac{x}{y}, s = \frac{y}{z}, t = \frac{z}{x}$. Prove that

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0.$$

Solution:

Given that $u = f(r, s, t)$

$$r = \frac{x}{y}; s = \frac{y}{z}; t = \frac{z}{x}$$

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial x} + \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial x} + \frac{\partial u}{\partial t} \cdot \frac{\partial t}{\partial x} = \frac{\partial u}{\partial r} \cdot \frac{1}{y} + \frac{\partial u}{\partial s}(0) + \frac{\partial u}{\partial t} \left(\frac{-z}{x^2} \right) \\ x \frac{\partial u}{\partial x} &= \frac{x}{y} \left(\frac{\partial u}{\partial r} \right) - \frac{z}{x} \left(\frac{\partial u}{\partial t} \right) \end{aligned} \quad \dots(1)$$

$$\begin{aligned} \frac{\partial u}{\partial y} &= \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial y} + \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial y} + \frac{\partial u}{\partial t} \cdot \frac{\partial t}{\partial y} = \frac{\partial u}{\partial r} \left(\frac{-x}{y^2} \right) + \frac{\partial u}{\partial s} \left(\frac{1}{z} \right) + \frac{\partial u}{\partial t}(0) \\ y \frac{\partial u}{\partial y} &= \left(\frac{-x}{y} \right) \left(\frac{\partial u}{\partial r} \right) + \frac{y}{z} \left(\frac{\partial u}{\partial s} \right) \end{aligned} \quad \dots(2)$$

$$\text{Similarly, } z \frac{\partial u}{\partial z} = \left(\frac{-y}{z} \right) \left(\frac{\partial u}{\partial s} \right) + \frac{z}{x} \left(\frac{\partial u}{\partial t} \right) \quad \dots(3)$$

Adding equations (1), (2) and (3) we get,

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = \frac{x}{y} \left(\frac{\partial u}{\partial r} \right) - \frac{z}{x} \left(\frac{\partial u}{\partial t} \right) - \frac{x}{y} \left(\frac{\partial u}{\partial r} \right) + \frac{y}{z} \left(\frac{\partial u}{\partial s} \right) - \frac{y}{z} \left(\frac{\partial u}{\partial s} \right) + \frac{z}{x} \left(\frac{\partial u}{\partial t} \right)$$

$$\text{(i. e.) } x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$$

Example 7: Transfer the equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ into polar co-ordinates.

Solution:

We know that the relation between the cartesian (x, y) and polar co-ordinates (r, θ) are

$$x = r \cos \theta; y = r \sin \theta$$

$$\text{Also, } x^2 + y^2 = r^2 \Rightarrow r = \sqrt{x^2 + y^2}, \quad \tan \theta = \frac{y}{x} \Rightarrow \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$$\frac{\partial r}{\partial x} = \frac{x}{\sqrt{x^2 + y^2}} = \frac{x}{r} = \cos \theta; \quad \frac{\partial r}{\partial y} = \frac{y}{\sqrt{x^2 + y^2}} = \frac{y}{r} = \sin \theta$$

$$\frac{\partial \theta}{\partial x} = \frac{1}{1 + \cancel{y^2/x^2}} \left(\frac{-y}{x^2} \right) = \frac{-y}{x^2 + y^2} = \frac{-r \sin \theta}{r^2} = \frac{-\sin \theta}{r}$$

$$\frac{\partial \theta}{\partial y} = \frac{1}{1 + \cancel{y^2/x^2}} \left(\frac{1}{x} \right) = \frac{x}{x^2 + y^2} = \frac{r \cos \theta}{r^2} = \frac{\cos \theta}{r}$$

Here u is a composite function of x and y .

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial x} + \frac{\partial u}{\partial \theta} \cdot \frac{\partial \theta}{\partial x} = \cos \theta \frac{\partial u}{\partial r} - \frac{\sin \theta}{r} \frac{\partial u}{\partial \theta} \\ \frac{\partial(u)}{\partial x} &= \left(\cos \theta \frac{\partial}{\partial r} - \frac{\sin \theta}{r} \frac{\partial}{\partial \theta} \right) u \Rightarrow \frac{\partial}{\partial x} = \cos \theta \frac{\partial}{\partial r} - \frac{\sin \theta}{r} \frac{\partial}{\partial \theta} \quad \dots(1) \end{aligned}$$

Similarly,

$$\begin{aligned} \frac{\partial u}{\partial y} &= \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial y} + \frac{\partial u}{\partial \theta} \cdot \frac{\partial \theta}{\partial y} = \sin \theta \frac{\partial u}{\partial r} + \frac{\cos \theta}{r} \frac{\partial u}{\partial \theta} \\ \therefore \frac{\partial}{\partial y} &= \sin \theta \frac{\partial}{\partial r} + \frac{\cos \theta}{r} \frac{\partial}{\partial \theta} \\ \frac{\partial^2 u}{\partial x^2} &= \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right) = \left(\cos \theta \frac{\partial}{\partial r} - \frac{\sin \theta}{r} \frac{\partial}{\partial \theta} \right) \left(\cos \theta \frac{\partial u}{\partial r} - \frac{\sin \theta}{r} \frac{\partial u}{\partial \theta} \right) \\ &= \cos \theta \frac{\partial}{\partial r} \left(\cos \theta \frac{\partial u}{\partial r} - \frac{\sin \theta}{r} \frac{\partial u}{\partial \theta} \right) - \frac{\sin \theta}{r} \frac{\partial}{\partial \theta} \left(\cos \theta \frac{\partial u}{\partial r} - \frac{\sin \theta}{r} \frac{\partial u}{\partial \theta} \right) \\ &= \left[\cos^2 \theta \frac{\partial^2 u}{\partial r^2} - \sin \theta \cos \theta \frac{\partial u}{\partial \theta} \left(\frac{-1}{r^2} \right) - \frac{\sin \theta}{r} \cdot \frac{\partial^2 u}{\partial r \partial \theta} \right] \\ &\quad + \left[\frac{\sin^2 \theta}{r} \cdot \frac{\partial u}{\partial r} - \frac{\sin \theta \cos \theta}{r} \frac{\partial^2 u}{\partial \theta \partial r} + \frac{\sin \theta \cos \theta}{r^2} \frac{\partial u}{\partial \theta} + \frac{\sin^2 \theta}{r^2} \cdot \frac{\partial^2 u}{\partial \theta^2} \right] \end{aligned}$$

$$\frac{\partial^2 u}{\partial x^2} = \cos^2 \theta \frac{\partial^2 u}{\partial r^2} + \frac{2 \sin \theta \cos \theta}{r^2} \frac{\partial u}{\partial \theta} + \frac{\sin^2 \theta}{r} \cdot \frac{\partial u}{\partial r} - \frac{2 \sin \theta \cos \theta}{r} \frac{\partial^2 u}{\partial r \partial \theta} + \frac{\sin^2 \theta}{r^2} \cdot \frac{\partial^2 u}{\partial \theta^2} \quad \dots(3)$$

$$\begin{aligned} \frac{\partial^2 u}{\partial y^2} &= \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial y} \right) = \left(\sin \theta \frac{\partial}{\partial r} + \frac{\cos \theta}{r} \frac{\partial}{\partial \theta} \right) \left(\sin \theta \frac{\partial u}{\partial r} + \frac{\cos \theta}{r} \frac{\partial u}{\partial \theta} \right) \\ &= \sin \theta \frac{\partial}{\partial r} \left(\sin \theta \frac{\partial u}{\partial r} + \frac{\cos \theta}{r} \frac{\partial u}{\partial \theta} \right) + \frac{\cos \theta}{r} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial r} + \frac{\cos \theta}{r} \frac{\partial u}{\partial \theta} \right) \\ &= \sin^2 \theta \frac{\partial^2 u}{\partial r^2} - \frac{2 \sin \theta \cos \theta}{r^2} \frac{\partial u}{\partial \theta} + \frac{\cos^2 \theta}{r} \cdot \frac{\partial u}{\partial r} + \frac{2 \cos \theta \sin \theta}{r} \frac{\partial^2 u}{\partial r \partial \theta} + \frac{\cos^2 \theta}{r^2} \cdot \frac{\partial^2 u}{\partial \theta^2} \quad \dots(4) \end{aligned}$$

Adding (3) and (4), we get

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} &= (\cos^2 \theta + \sin^2 \theta) \frac{\partial^2 u}{\partial r^2} + \frac{(\sin^2 \theta + \cos^2 \theta)}{r} \frac{\partial u}{\partial r} + \left(\frac{\sin^2 \theta + \cos^2 \theta}{r^2} \right) \cdot \frac{\partial^2 u}{\partial \theta^2} \\ &= \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} \end{aligned}$$

$$\therefore \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \text{is transformed into} \quad \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0.$$

Exercises

1. P.T. $u \frac{\partial z}{\partial u} - v \frac{\partial z}{\partial v} = 2(x^2 - y^2)^{-1/2} \frac{\partial z}{\partial x}$ if $z = f(x, y)$ and $x = u^2 + v^2$ and $y = 2uv$
2. P.T. $(y^2 - zx) \frac{\partial u}{\partial x} + (x^2 - yz) \frac{\partial u}{\partial y} + (z^2 - xy) \frac{\partial u}{\partial z} = 0$ if $u = f(x^2, 2yz, y^2 + 2xz)$
3. P.T. $\left(\frac{\partial z}{\partial u} \right)^2 + \left(\frac{\partial z}{\partial v} \right)^2 = e^{2u} \left[\left(\frac{\partial z}{\partial x} \right)^2 + \left(\frac{\partial z}{\partial y} \right)^2 \right]$ if $z = f(x, y)$ and $x = e^u \cos v$,
 $y = e^u \sin v$
4. If $u = f\left[\frac{y-x}{xy}, \frac{z-x}{xz}\right]$, Show that $x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} + z^2 \frac{\partial u}{\partial z} = 0$.
5. P.T. $\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y}$ if $z = f(x, y)$ where $x = e^u + e^{-v}$ and $y = e^{-u} - e^v$

6. P.T. $\frac{1}{x} \frac{\partial u}{\partial x} + \frac{1}{y} \frac{\partial u}{\partial y} + \frac{1}{z} \frac{\partial u}{\partial z} = 0$ if $u = f(x^2 - y^2, y^2 - z^2, z^2 - x^2)$.
7. P.T. $\left(\frac{\partial v}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial y}\right)^2 = \left(\frac{\partial v}{\partial \rho}\right)^2 + \frac{1}{\rho^2} \left(\frac{\partial v}{\partial f}\right)^2$ if v is a function of x and y and $x = \rho \cos \phi, y = \rho \sin \phi$
8. P.T. $a \frac{\partial z}{\partial x} + b \frac{\partial z}{\partial y} = c$ if $f(cx - az, cy - bz) = 0$

2.9 JACOBIANS

If u and v are functions of two independent variables x and y , then the determinant

$\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$ is called the Jacobian or functional determinant of u and v with respect to x and

y and is denoted by $\frac{\partial(u, v)}{\partial(x, y)}$ or $J\left(\frac{u, v}{x, y}\right)$.

Similarly, if u, v, w are functions of three independent variables x, y and z , then the

determinant
$$\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$$
 is called the Jacobian of u, v, w with respect to x, y, z and is denoted by $\frac{\partial(u, v, w)}{\partial(x, y, z)}$ or $J\left(\frac{u, v, w}{x, y, z}\right)$.

Note: If the functions u, v, w of three independent variables x, y, z are not independent, then

the Jacobian of u, v, w with respect to x, y, z vanishes. (i.e.) $\frac{\partial(u, v, w)}{\partial(x, y, z)} = 0$

2.9.1 Properties of Jacobian

1. If u and v are functions of x and y , then $\frac{\partial(u, v)}{\partial(x, y)} \cdot \frac{\partial(x, y)}{\partial(u, v)} = 1$ (or) $J \cdot J' = 1$.

$$\text{Proof: } J \cdot J' = \frac{\partial(u, v)}{\partial(x, y)} \cdot \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} \times \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

By interchanging rows and columns of the second determinant.

$$= \begin{vmatrix} \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial u} & \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial v} \\ \frac{\partial v}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial v}{\partial y} \cdot \frac{\partial y}{\partial u} & \frac{\partial v}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial v}{\partial y} \cdot \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1.$$

$$\therefore JJ' = 1$$

2. If u and v are functions of r and s where r and s are functions of x and y then

$$\frac{\partial(u, v)}{\partial(x, y)} = \frac{\partial(u, v)}{\partial(r, s)} \times \frac{\partial(r, s)}{\partial(x, y)}.$$

Proof:

$$\begin{aligned}
 \frac{\partial(u,v)}{\partial(r,s)} \times \frac{\partial(r,s)}{\partial(x,y)} &= \begin{vmatrix} \frac{\partial u}{\partial r} & \frac{\partial u}{\partial s} \\ \frac{\partial v}{\partial r} & \frac{\partial v}{\partial s} \end{vmatrix} \begin{vmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} \\ \frac{\partial s}{\partial x} & \frac{\partial s}{\partial y} \end{vmatrix} = \begin{vmatrix} \frac{\partial u}{\partial r} & \frac{\partial u}{\partial s} \\ \frac{\partial v}{\partial r} & \frac{\partial v}{\partial s} \end{vmatrix} \begin{vmatrix} \frac{\partial r}{\partial x} & \frac{\partial s}{\partial x} \\ \frac{\partial r}{\partial y} & \frac{\partial s}{\partial y} \end{vmatrix} \\
 &= \begin{vmatrix} \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial x} + \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial x} & \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial y} + \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial y} \\ \frac{\partial v}{\partial r} \cdot \frac{\partial r}{\partial x} + \frac{\partial v}{\partial s} \cdot \frac{\partial s}{\partial x} & \frac{\partial v}{\partial r} \cdot \frac{\partial r}{\partial y} + \frac{\partial v}{\partial s} \cdot \frac{\partial s}{\partial y} \end{vmatrix} \\
 &= \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} \\
 &= \frac{\partial(u,v)}{\partial(x,y)}.
 \end{aligned}$$

Hence proved.

Points to Remember:

1. Jacobian of two functions $u(x,y)$ and $v(x,y)$ where x and y are

independent variables: $J\left(\frac{u,v}{x,y}\right) = \frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}.$

2. Jacobian of three functions $u(x,y,z)$, $v(x,y,z)$ and $w(x,y,z)$ where x,y,z

are independent variables: $J\left(\frac{u,v,w}{x,y,z}\right) = \frac{\partial(u,v,w)}{\partial(x,y,z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}.$

3. If u and v are functions of x and y , then $\frac{\partial(u,v)}{\partial(x,y)} \cdot \frac{\partial(x,y)}{\partial(u,v)} = 1$ (or) $J \cdot J' = 1$.

4. If u and v are functions of r and s where r and s are functions of x and y then $\frac{\partial(u,v)}{\partial(x,y)} = \frac{\partial(u,v)}{\partial(r,s)} \times \frac{\partial(r,s)}{\partial(x,y)}.$

Example 1: If $x = r \cos \theta$, $y = r \sin \theta$, prove that $\frac{\partial(x, y)}{\partial(r, \theta)} \times \frac{\partial(r, \theta)}{\partial(x, y)} = 1$.

Solution:

Given that $x = r \cos \theta$; $y = r \sin \theta$

$$r^2 = x^2 + y^2 \quad ; \quad \tan \theta = \frac{y}{x}$$

$$r = \sqrt{x^2 + y^2} \quad ; \quad \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$$\therefore r^2 = x^2 + y^2$$

$$2r \frac{\partial r}{\partial x} = 2x \quad \Rightarrow \quad \frac{\partial r}{\partial x} = \frac{x}{r}$$

Similarly, $\frac{\partial r}{\partial y} = \frac{y}{r}$

Similarly, $\frac{\partial \theta}{\partial x} = \frac{1}{1 + \cancel{y^2} / x^2} \cdot \frac{-y}{x^2} = \frac{-y}{x^2 + y^2} = \frac{-y}{r^2}$ and $\frac{\partial \theta}{\partial y} = \frac{x}{r^2}$

$$\frac{\partial(r, \theta)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} \\ \frac{\partial \theta}{\partial x} & \frac{\partial \theta}{\partial y} \end{vmatrix} = \begin{vmatrix} \frac{x}{r} & \frac{y}{r} \\ \frac{-y}{r^2} & \frac{x}{r^2} \end{vmatrix} = \frac{x^2}{r^3} + \frac{y^2}{r^3}$$

$$\begin{aligned}
 &= \frac{x^2 + y^2}{r^3} = \frac{r}{r^3} = \frac{1}{r} \frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} \\
 &= r \cos^2 \theta + r \sin^2 \theta = r
 \end{aligned}$$

From the above equation

$$\frac{\partial(r, \theta)}{\partial(x, y)} \cdot \frac{\partial(x, y)}{\partial(r, \theta)} = r \left(\frac{1}{r} \right) = 1.$$

Example 2: If $x = uv$; $y = \frac{u}{v}$ prove that $\frac{\partial(x, y)}{\partial(u, v)} \cdot \frac{\partial(u, v)}{\partial(x, y)} = 1$.

Solution:

Given that $x = uv$; $y = \frac{u}{v}$

$$\Rightarrow \frac{\partial x}{\partial u} = v ; \frac{\partial x}{\partial v} = u$$

$$xy = u^2 \Rightarrow u = \sqrt{xy}$$

$$\Rightarrow \frac{\partial y}{\partial u} = \frac{1}{v} ; \frac{\partial y}{\partial v} = \frac{-u}{v^2}$$

$$\frac{x}{y} = \frac{1}{v^2} ; v = \sqrt{\frac{x}{y}}$$

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} v & u \\ \frac{1}{v} & \frac{-u}{v^2} \end{vmatrix} = -\frac{2u}{v}$$

$$J' = \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} \frac{\sqrt{y}}{2\sqrt{x}} & \frac{\sqrt{x}}{2\sqrt{y}} \\ \frac{1}{2\sqrt{xy}} & \frac{-\sqrt{x}}{2\sqrt{y} \cdot y} \end{vmatrix}$$

$$= \begin{vmatrix} \frac{y}{2u} & \frac{x}{2u} \\ \frac{1}{2vy} & \frac{-x}{2vy^2} \end{vmatrix} = \frac{-x}{4uvy} - \frac{x}{4uvy} = \frac{-x}{2uvy}$$

$$JJ' = \left(\frac{-2u}{v} \right) \left(\frac{-x}{2uvy} \right) = \frac{x}{v^2 y} = 1 \quad \left[\because v = \sqrt{\frac{x}{y}} \right]$$

$$JJ' = 1$$

Example 3: If $u = 2xy, v = x^2 - y^2, x = r \cos \theta, y = r \sin \theta$. Evaluate $\frac{\partial(u, v)}{\partial(r, \theta)}$.

Solution:

$$\begin{aligned} \frac{\partial(u, v)}{\partial(r, \theta)} &= \frac{\partial(u, v)}{\partial(x, y)} \cdot \frac{\partial(x, y)}{\partial(r, \theta)} \\ &= \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} \cdot \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} 2y & 2x \\ 2x & -2y \end{vmatrix} \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} \\ &= (-4y^2 - 4x^2)(r \cos^2 \theta + r \sin^2 \theta) = -4(x^2 + y^2)r \\ \frac{\partial(u, v)}{\partial(r, \theta)} &= -4r^3. \end{aligned}$$

Example 4: If $p = 3x + 2y - z, q = x - 2y + z, r = x + 2y - z$. Prove that

$$\frac{\partial(p, q, r)}{\partial(x, y, z)} = 0.$$

Solution:

$$\text{Consider } \frac{\partial(p, q, r)}{\partial(x, y, z)} = \begin{vmatrix} \frac{\partial p}{\partial x} & \frac{\partial p}{\partial y} & \frac{\partial p}{\partial z} \\ \frac{\partial q}{\partial x} & \frac{\partial q}{\partial y} & \frac{\partial q}{\partial z} \\ \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} & \frac{\partial r}{\partial z} \end{vmatrix} = \begin{vmatrix} 3 & 2 & -1 \\ 1 & -2 & 1 \\ 1 & 2 & -1 \end{vmatrix} = -2 \begin{vmatrix} 3 & -1 & -1 \\ 1 & 1 & 1 \\ 1 & -1 & -1 \end{vmatrix}$$

$$\therefore \frac{\partial(p, q, r)}{\partial(x, y, z)} = 0$$

Example 5: If $x = u(1-v)$, $y = uv$, compute the Jacobians $J = \frac{\partial(x, y)}{\partial(u, v)}$ and

$J' = \frac{\partial(u, v)}{\partial(x, y)}$ and verify that $JJ' = 1$.

Solution:

$$\text{Given that } x = u(1-v) = u - uv \quad \dots(1)$$

$$y = uv \quad \dots(2)$$

$$\frac{\partial x}{\partial u} = 1-v, \quad \frac{\partial x}{\partial v} = 0-u = -u$$

$$\frac{\partial y}{\partial u} = v, \quad \frac{\partial y}{\partial v} = u$$

$$\therefore J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 1-v & -u \\ v & u \end{vmatrix} = u(1-v) + uv = u$$

$$J = u$$

To find J' :

$$\text{From equation (1) and (2) } x = u - y \quad y = uv$$

$$\begin{array}{l|l}
 \therefore u = x + y & v = \frac{y}{u} = \frac{y}{x + y} \\
 \frac{\partial u}{\partial x} = 1 & \frac{\partial v}{\partial x} = y \cdot \frac{-1}{(x + y)^2} = \frac{-y}{(x + y)^2} \\
 \frac{\partial u}{\partial y} = 1 & \frac{\partial v}{\partial y} = \frac{(x + y)(1) - y(1)}{(x + y)^2} = \frac{x}{(x + y)^2}
 \end{array}$$

$$J' = \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ \frac{-y}{(x + y)^2} & \frac{x}{(x + y)^2} \end{vmatrix}$$

$$= \frac{x}{(x + y)^2} + \frac{y}{(x + y)^2} = \frac{1}{x + y}$$

$$J' = \frac{1}{u}$$

Hence, $JJ' = u \times \frac{1}{u} = 1$

Example 6: If $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$, $z = r \cos \theta$, prove that

$$\frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} = r^2 \sin \theta.$$

Solution:

Given that $x = r \sin \theta \cos \phi$

$$\frac{\partial x}{\partial r} = \sin \theta \cos \phi; \frac{\partial x}{\partial \theta} = r \cos \theta \cos \phi; \frac{\partial x}{\partial \phi} = -r \sin \theta \sin \phi$$

$$\frac{\partial y}{\partial r} = \sin \theta \sin \phi; \frac{\partial y}{\partial \theta} = r \cos \theta \sin \phi; \frac{\partial y}{\partial \phi} = r \sin \theta \cos \phi$$

$$\frac{\partial z}{\partial r} = \cos \theta ; \frac{\partial z}{\partial \theta} = -r \sin \theta ; \frac{\partial z}{\partial \phi} = 0$$

$$\begin{aligned} \therefore \frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} &= \begin{vmatrix} \sin \theta \cos \phi & r \cos \theta \cos \phi & -r \sin \theta \sin \phi \\ \sin \theta \sin \phi & r \cos \theta \sin \phi & r \sin \theta \cos \phi \\ \cos \theta & -r \sin \theta & 0 \end{vmatrix} \\ &= -r \sin \theta \sin \phi (-r \sin^2 \theta \sin \phi - r \cos^2 \theta \sin \phi) \\ &\quad - r \sin \theta \cos \phi (-r \sin^2 \theta \cos \phi - r \cos^2 \theta \cos \phi) \\ &= r^2 \sin \theta \sin^2 \phi + r^2 \sin \theta \cos^2 \phi = r^2 \sin \theta (\sin^2 \phi + \cos^2 \phi) \\ \frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} &= r^2 \sin \theta \end{aligned}$$

Example 7: If $x + y + z = u$; $y + z = uv$; $z = uvw$, prove $\frac{\partial(x, y, z)}{\partial(u, v, w)} = u^2 v$.

Solution:

Given that $z = uvw$

$$y = uv - z = uv - uvw, \quad y = uv - z$$

$$x = u - (y + z) = u - uv$$

$$x = u(1 - v)$$

$$y = uv(1 - w)$$

$$z = uvw$$

$$\frac{\partial x}{\partial u} = 1 - v$$

$$\frac{\partial y}{\partial u} = v(1 - w)$$

$$\frac{\partial z}{\partial u} = vw$$

$$\frac{\partial x}{\partial v} = -u$$

$$\frac{\partial y}{\partial v} = u(1 - w)$$

$$\frac{\partial z}{\partial v} = uw$$

$$\frac{\partial x}{\partial w} = 0$$

$$\frac{\partial y}{\partial w} = -uv$$

$$\frac{\partial z}{\partial w} = uv$$

$$\begin{aligned}
\therefore \frac{\partial(x, y, z)}{\partial(u, v, w)} &= \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix} = \begin{vmatrix} 1-v & -u & 0 \\ v(1-w) & u(1-w) & -uv \\ vw & uw & uv \end{vmatrix} \\
&= (1-v) \{ [u(1-w) \cdot uv] + u^2vw \} + u \{ [v(1-w) \cdot uv] + uv^2w \} \\
&= (1-v) [u^2v(1-w) + u^2vw] + u [uv^2(1-w) + uv^2w] \\
&= (1-v) u^2v [1-w+w] + u^2v^2 [1-w+w] \\
&= (1-v) u^2v + u^2v^2 = u^2v - u^2v^2 + u^2v^2 = u^2v \\
\therefore \frac{\partial(x, y, z)}{\partial(u, v, w)} &= u^2v
\end{aligned}$$

Example 8: If $x = e^r \sec \theta$; $y = e^r \tan \theta$, prove that $\frac{\partial(x, y)}{\partial(r, \theta)} \cdot \frac{\partial(r, \theta)}{\partial(x, y)} = 1$.

Solution:

To find J :

$$\begin{array}{l|l}
\text{Given that } x = e^r \sec \theta \dots(1) & y = e^r \tan \theta \dots(2) \\
\frac{\partial x}{\partial r} = e^r \sec \theta & \frac{\partial y}{\partial r} = e^r \tan \theta \\
\frac{\partial x}{\partial \theta} = e^r \sec \theta \tan \theta & \frac{\partial y}{\partial \theta} = e^r \sec^2 \theta
\end{array}$$

$$\begin{aligned}
J = \frac{\partial(x, y)}{\partial(r, \theta)} &= \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial y}{\partial r} \\ \frac{\partial x}{\partial \theta} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} e^r \sec \theta & e^r \tan \theta \\ e^r \sec \theta \tan \theta & e^r \sec^2 \theta \end{vmatrix} \\
&= e^{2r} \sec^3 \theta - e^{2r} \sec \theta \tan^2 \theta
\end{aligned}$$

$$= e^{2r} \sec \theta (\sec^2 \theta - \tan^2 \theta) \quad \because \sec^2 \theta - \tan^2 \theta = 1$$

$$J = e^{2r} \sec \theta$$

To find J' :

From equations (1) and (2), $x^2 - y^2 = e^{2r} \sec^2 \theta - e^{2r} \tan^2 \theta = e^{2r}$

$$x^2 - y^2 = e^{2r} \Rightarrow 2r = \log(x^2 - y^2)$$

$$\therefore r = \frac{1}{2} \log(x^2 - y^2)$$

Also $\frac{y}{x} = \frac{e^r \tan \theta}{e^r \sec \theta} = \sin \theta$

$$\frac{y}{x} = \sin \theta \Rightarrow \theta = \sin^{-1}\left(\frac{y}{x}\right)$$

$$J' = \frac{\partial(r, \theta)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} \\ \frac{\partial \theta}{\partial x} & \frac{\partial \theta}{\partial y} \end{vmatrix}$$

$$\because r = \frac{1}{2} \log(x^2 - y^2) \quad \text{and} \quad \theta = \sin^{-1}\left(\frac{y}{x}\right)$$

$$\frac{\partial r}{\partial x} = \frac{1}{2} \frac{1}{x^2 - y^2} (2x) = \frac{x}{x^2 - y^2}$$

$$\frac{\partial r}{\partial y} = \frac{1}{2} \frac{1}{x^2 - y^2} (-2y) = \frac{-y}{x^2 - y^2}$$

$$\frac{\partial \theta}{\partial x} = \frac{1}{\sqrt{1 - \left(\frac{y}{x}\right)^2}} \cdot \left(-\frac{y}{x^2}\right) = \frac{-y}{x^2} \frac{x}{\sqrt{x^2 - y^2}} = \frac{-y}{x\sqrt{x^2 - y^2}}$$

$$\frac{\partial \theta}{\partial y} = \frac{1}{\sqrt{1 - \left(\frac{y}{x}\right)^2}} \cdot \left(\frac{1}{x}\right) = \frac{1}{x} \frac{x}{\sqrt{x^2 - y^2}} = \frac{1}{\sqrt{x^2 - y^2}}$$

$$\begin{aligned} J' &= \begin{vmatrix} \frac{x}{x^2 - y^2} & \frac{-y}{x^2 - y^2} \\ \frac{-y}{x\sqrt{x^2 - y^2}} & \frac{1}{\sqrt{x^2 - y^2}} \end{vmatrix} = \frac{x}{(x^2 - y^2)\sqrt{x^2 - y^2}} - \frac{y^2}{x(x^2 - y^2)\sqrt{x^2 - y^2}} \\ &= \frac{1}{(x^2 - y^2)\sqrt{x^2 - y^2}} \left[x - \frac{y^2}{x} \right] = \frac{(x^2 - y^2)}{x(x^2 - y^2)\sqrt{x^2 - y^2}} \\ &= \frac{1}{x\sqrt{x^2 - y^2}} = \frac{1}{(e^r \sec \theta)e^r} = \frac{1}{e^{2r} \sec \theta} \end{aligned}$$

$$\therefore JJ' = e^{2r} \sec \theta \cdot \frac{1}{e^{2r} \sec \theta} = 1$$

Exercises

1. If $u = x^2$ and $v = y^2$, find $\frac{\partial(u, v)}{\partial(x, y)}$.
2. If $x = r \cos \theta$; $y = r \sin \theta$; $z = z$, find $\frac{\partial(x, y, z)}{\partial(r, \theta, z)}$.
3. If $x = u(1-v)$; $y = uv$, prove that $\frac{\partial(x, y)}{\partial(u, v)} \cdot \frac{\partial(u, v)}{\partial(x, y)} = 1$
4. If $u = xyz$, $v = xy + yz + zx$ and $w = x + y + z$, find $\frac{\partial(u, v, w)}{\partial(x, y, z)}$.
5. If $y_1 = \frac{x_2 x_3}{x_1}$, $y_2 = \frac{x_3 x_1}{x_2}$, $y_3 = \frac{x_1 x_2}{x_3}$, prove that the Jacobian of y_1, y_2, y_3 with respect to x_1, x_2, x_3 is 4.
6. If $x = a \cosh u \cos v$; $y = a \sinh u \sin v$, prove that $\frac{\partial(x, y)}{\partial(u, v)} = \frac{a^2}{2} (\cosh 2u - \cos 2v)$.

7. P.T. $J \frac{(x, y, z)}{(u, v, w)} = -\sin^3 u \sin^2 v \sin w$ if $x = \cos u$; $y = \cos u \sin u$;
 $z = \cos w \sin v \sin u$.
8. If $u = x(1-r^2)^{-1/2}$, $v = y(1-r^2)^{-1/2}$, $w = z(1-r^2)^{-1/2}$ where $r^2 = x^2 + y^2 + z^2$.
 Prove that $\frac{\partial(u, v, w)}{\partial(x, y, z)} = (1-r^2)^{-5/2}$.

Answers

1. $4xy$.
2. r
3. $(x-y)(y-z)(z-x)$.

2.10 TAYLOR'S SERIES FOR FUNCTIONS OF TWO VARIABLES

Taylor's series for a function of single variable x is defined as

$$f(x+h) = f(x) + \frac{h}{1!} f'(x) + \frac{h^2}{2!} f''(x) + \dots$$

Let us consider a function of $f(x, y)$ of two independent variables x and y . Expanding using Taylor's series for a function of single variable x by treating y as constant, we get,

$$f(x+h, y+k) = f(x, y+k) + h \frac{\partial}{\partial x} f(x, y+k) + \frac{h^2}{2!} \frac{\partial^2}{\partial x^2} f(x, y+k) + \dots \quad \dots(1)$$

Similarly treating x as constant, apply Taylor's series for a function of a single variable y , we get,

$$f(x, y+k) = f(x, y) + \frac{k}{1!} \frac{\partial}{\partial y} f(x, y) + \frac{k^2}{2!} \frac{\partial^2}{\partial y^2} f(x, y) + \dots \quad \dots(2)$$

Using equation (2) in (1), we get

$$f(x+h, y+k) = \left[f(x, y) + \frac{k}{1!} \frac{\partial}{\partial y} f(x, y) + \frac{k^2}{2!} \frac{\partial^2}{\partial y^2} f(x, y) + \dots \right]$$

$$\begin{aligned}
& + \frac{h}{1!} \frac{\partial}{\partial x} \left[f(x, y) + \frac{k}{1!} \frac{\partial}{\partial y} f(x, y) + \dots \right] + \frac{h^2}{2!} \frac{\partial^2}{\partial x^2} \left[f(x, y) + k \frac{\partial}{\partial y} f(x, y) + \dots \right] \\
& = \left[f(x, y) + k \frac{\partial f}{\partial y} + \frac{k^2}{2!} \frac{\partial^2 f}{\partial y^2} + \dots \right] + \left[h \frac{\partial f}{\partial x} + hk \frac{\partial^2 f}{\partial x \partial y} + hk^2 \frac{\partial^2 f}{\partial x \partial y^2} + \dots \right] \\
& \quad + \left[\frac{h^2}{2!} \frac{\partial^2 f}{\partial x^2} + \frac{h^2 k}{2!} \frac{\partial^3 f}{\partial x^2 \partial y} + \dots \right] \\
& = f(x, y) + \frac{1}{1!} \left[h \frac{\partial f}{\partial x} + k \frac{\partial f}{\partial y} \right] + \left[\frac{h^2}{2!} \frac{\partial^2 f}{\partial x^2} + hk \frac{\partial^2 f}{\partial x \partial y} + \frac{k^2}{2!} \frac{\partial^2 f}{\partial y^2} \right] \\
& \quad + \left[\frac{h^3}{3!} \frac{\partial^3 f}{\partial x^3} + \frac{h^2 k}{2!} \frac{\partial^3 f}{\partial x^2 \partial y} + \frac{hk^2}{2!} \frac{\partial^3 f}{\partial x \partial y^2} + \frac{k^3}{3!} \frac{\partial^3 f}{\partial y^3} \right] + \dots \\
& = f(x, y) + \left[h \frac{\partial f}{\partial x} + k \frac{\partial f}{\partial y} \right] + \frac{1}{2!} \left[h^2 \frac{\partial^2 f}{\partial x^2} + 2hk \frac{\partial^2 f}{\partial x \partial y} + k^2 \frac{\partial^2 f}{\partial y^2} \right] \\
& \quad + \frac{1}{3!} \left[h^3 \frac{\partial^3 f}{\partial x^3} + 3h^2 k \frac{\partial^3 f}{\partial x^2 \partial y} + 3hk^2 \frac{\partial^3 f}{\partial x \partial y^2} + k^3 \frac{\partial^3 f}{\partial y^3} \right] + \dots \\
& = f(x, y) + \frac{1}{1!} \left[h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right] f + \frac{1}{2!} \left[h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right]^2 f + \frac{1}{3!} \left[h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right]^3 f + \dots \\
& \quad \dots (3)
\end{aligned}$$

Corollary 1: Putting $x = a$ and $y = b$ in equation (3) we get,

$$\begin{aligned}
f(a+h, b+k) &= f(a, b) + \left[hf_x(a, b) + kf_y(a, b) \right] \\
& + \frac{1}{2!} \left[h^2 f_{xx}(a, b) + 2hk f_{xy}(a, b) + k^2 f_{yy}(a, b) \right] \\
& + \frac{1}{3!} \left[h^3 f_{xxx}(a, b) + 2h^2 k f_{xxy}(a, b) + 3hk^2 f_{xyy}(a, b) + h^3 f_{yyy}(a, b) \right] + \dots
\end{aligned}$$

Corollary 2: Putting $a+h = x$, $b+k = y$, then $h = x-a$, $k = y-b$, we get

$$f(x, y) = f(a, b) + [(x-a)f_x(a, b) + (y-b)f_y(a, b)] \\ + \frac{1}{2!}[(x-a)^2 f_{xx}(a, b) + 2(x-a)(y-b)f_{xy}(a, b) + (y-b)^2 f_{yy}(a, b)] + \dots \quad \dots(4)$$

This is used to expand $f(x, y)$ in the neighborhood of (a, b) .

Corollary 3: Putting $a = 0$, $b = 0$ in Corollary 2, we have

$$f(x, y) = f(0, 0) + [xf_x(0, 0) + yf_y(0, 0)] \\ + \frac{1}{2!}[x^2 f_{xx}(0, 0) + 2xyf_{xy}(0, 0) + y^2 f_{yy}(0, 0)] + \dots$$

This is called Maclaurin's series (or) Maclaurin's theorem for two variables.

This is used to expand $f(x, y)$ in powers of x and y or in the neighbourhood of the origin $(0, 0)$.

Note: The expansion (4) is called expansion of $f(x, y)$ at (a, b) or in the neighborhood of (a, b) or in powers of $(x-a)$ and $(y-b)$.

Taylor's series for a Function of Two Variables:

$$f(x+h, y+k) = f(x, y) + \frac{1}{1!}\left[h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right]f + \frac{1}{2!}\left[h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right]^2 f \\ + \frac{1}{3!}\left[h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right]^3 f + \dots$$

Example 1 : Expand $e^x \cos y$ in powers of x & y up to the second degree terms.

Solution:

$f(x, y) = e^x \cos y$	$f(0, 0) = 1$
$f_x = e^x \cos y$	$f_x(0, 0) = 1$

$$\begin{aligned}
 f_{xx} &= e^x \cos y & f_{xx}(0,0) &= 1 \\
 f_y &= e^x (-\sin y) & f_y(0,0) &= 0 \\
 f_{yy} &= e^x (-\cos y) & f_{yy}(0,0) &= -1 \\
 f_{xy} &= -e^x \sin y & f_{xy}(0,0) &= 0
 \end{aligned}$$

Taylor's series for $f(x, y)$ in powers of x and y is

$$\begin{aligned}
 f(x, y) &= f(0,0) + \frac{1}{1!} [xf_x(0,0) + yf_y(0,0)] \\
 &\quad + \frac{1}{2!} [x^2 f_{xx}(0,0) + 2xyf_{xy}(0,0) + y^2 f_{yy}(0,0)] + \dots \\
 \therefore e^x \cos y &= 1 + \frac{1}{1!} [x(1) + y(0)] + \frac{1}{2!} [x^2(1) + 2xy(0) + y^2(-1)] \\
 &\quad + \frac{1}{3!} [x^3(1) + 3x^2y(0) + 3xy^2(-1) + y^3(0)] + \dots \\
 &= 1 + \frac{x}{1!} + \frac{1}{2!} [x^2 - y^2] + \dots \\
 \therefore e^x \cos y &= 1 + x + \frac{x^2}{2!} - \frac{y^2}{2!} + \dots
 \end{aligned}$$

Example 2: Find the Taylor series expansion of x^y near the point $(1,1)$ up to the second degree terms.

Solution:

Taylor's series of $f(x, y)$ near the point $(1,1)$ is $f(x, y)$ is given by

$$\begin{aligned}
 f(x, y) &= f(1,1) + \frac{1}{1!} [(x-1)f_x(1,1) + (y-1)f_y(1,1)] \\
 &\quad + \frac{1}{2!} [(x-1)^2 f_{xx}(1,1) + 2(x-1)(y-1)f_{xy}(1,1) + (y-1)^2 f_{yy}(1,1)] + \dots
 \end{aligned}$$

$$\begin{array}{l|l} f(x, y) = x^y & f(1, 1) = 1 \\ f_x = yx^{y-1} & f_x(1, 1) = 1 \\ f_{xx} = y(y-1)x^{y-2} & f_{xx}(1, 1) = 0 \\ f_y = x^y \log x & f_y(1, 1) = 0 \end{array}$$

$$f_{yy} = (\log x)x^y \log x = x^y (\log x)^2, \quad f_{yy}(1, 1) = 0$$

$$f_{xy} = (\log x)yx^{y-1} + x^y \left(\frac{1}{x}\right) = x^{y-1} + yx^{y-1} \log x, \quad f_{xy}(1, 1) = 1$$

$$\begin{aligned} \therefore x^y &= 1 + \frac{1}{1!}[(x-1)(1) + (y-1)(0)] + \frac{1}{2!}[(x-1)^2(0) + 2(x-1)(y-1)(1) + (y-1)^2(0)] \\ &= 1 + \frac{1}{1!}(x-1) + \frac{1}{2!}2(x-1)(y-1) \end{aligned}$$

$$x^y = 1 + (x-1) + (x-1)(y-1) + \dots$$

Example 3: Evaluate approximately $\log(\sqrt[3]{1.03} + \sqrt[4]{0.98} - 1)$.

Solution:

$$\text{Let } f(x+h, y+k) = \log\left[(x+h)^{1/3} + (y+k)^{1/4} - 1\right].$$

$$\text{Put } h = k = 0, \quad f(x, y) = \log\left(x^{1/3} + y^{1/4} - 1\right)$$

$$f_x = \frac{1}{x^{1/3} + y^{1/4} - 1} \left(\frac{1}{3} x^{-2/3} \right) = \frac{\frac{1}{3} x^{-2/3}}{x^{1/3} + y^{1/4} - 1}$$

$$f_y = \frac{1}{x^{1/3} + y^{1/4} - 1} \left(\frac{1}{4} x^{-3/4} \right) = \frac{\frac{1}{4} x^{-3/4}}{x^{1/3} + y^{1/4} - 1}$$

$$\text{Also, } \log\left[(x+h)^{1/3} + (y+k)^{1/4} - 1\right] = f(x+h, y+k)$$

$$= f(x, y) + hf_x + hf_y (app.,)$$

Put, $x = y = 1$, $h = 0.03$, $k = -0.02$, we get

$$\begin{aligned} \log(\sqrt[3]{1+1.03} + \sqrt[4]{1-0.02} - 1) &= f(1, 1) + 0.03f_x(1, 1) - 0.02f_y(1, 1) \\ &= \log(1+1-1) + 0.03\left(\frac{\frac{1}{3}(1)}{1+1-1}\right) - 0.02\left(\frac{\frac{1}{4}(1)}{1+1-1}\right) \\ &= \log 1 + 0.03\left(\frac{1}{3}\right) - 0.02\left(\frac{1}{4}\right) = 0 + 0.01 - 0.005 \end{aligned}$$

$$\therefore \log(\sqrt[3]{1+1.03} + \sqrt[4]{0.98} - 1) = 0.005 \text{ (app.,)}$$

Example 4: Expand $e^x \sin y$ in powers of x and y up to the third degree terms.

Solution:

$$\text{Let } f(x, y) = e^x \sin y \qquad f(0, 0) = 0$$

$$f_x = e^x \sin y \qquad f_x(0, 0) = 0$$

$$f_{xx} = e^x \sin y \qquad f_{xx}(0, 0) = 0$$

$$f_y = e^x \cos y \qquad f_y(0, 0) = 1$$

$$f_{yy} = -e^x \sin y \qquad f_{yy}(0, 0) = 0$$

$$f_{xy} = e^x \cos y \qquad f_{xy}(0, 0) = 1$$

$$f_{xxx} = e^x \sin y \qquad f_{xxx}(0, 0) = 0$$

$$f_{xxy} = e^x \cos y \qquad f_{xxy}(0, 0) = 1$$

$$f_{xyy} = -e^x \sin y \qquad f_{xyy}(0, 0) = 0$$

$$f_{yyy} = -e^x \cos y \qquad f_{yyy}(0, 0) = -1$$

Taylor series for $f(x, y)$ in powers of x and y is

$$\begin{aligned} f(x, y) = f(0, 0) + \frac{1}{1!} [xf_x(0, 0) + yf_y(0, 0)] \\ + \frac{1}{2!} [x^2 f_{xx}(0, 0) + 2xyf_{xy}(0, 0) + y^2 f_{yy}(0, 0)] \\ + \frac{1}{3!} [x^3 f_{xxx}(0, 0) + 3x^2 yf_{xxy}(0, 0) + 3xy^2 f_{xyy}(0, 0) + y^3 f_{yyy}(0, 0)] + \dots \end{aligned}$$

$$\begin{aligned} e^x \sin y = 0 + \frac{1}{1!} [x(0) + y(1)] + \frac{1}{2!} [x^2(0) + 2xy(1) + y^2(0)] \\ + \frac{1}{3!} [x^3(0) + 3x^2 y(1) + 3xy^2(0) + y^3(-1)] + \dots \end{aligned}$$

$$\therefore e^x \sin y = y + xy + \frac{1}{2} x^2 y - \frac{1}{6} y^3 + \dots$$

Example 5: Expand $x^2 y + 3y - 2$ in powers of $(x-1)$ and $(y+2)$ up to the third degree terms.

Solution:

$$\text{Let } f(x, y) = x^2 y + 3y - 2$$

The expansion of $f(x, y)$ in powers of $(x-a)$ & $(y-b)$ is given by Taylor's theorem,

$$\begin{aligned} \therefore f(x, y) = f(a, b) + [(x-a)f_x(a, b) + (y-b)f_y(a, b)] \\ + \frac{1}{2!} [(x-a)^2 f_{xx}(a, b) + 2(x-a)(y-b)f_{xy}(a, b) + (y-b)^2 f_{yy}(a, b)] \\ + \frac{1}{3!} [(x-a)^3 f_{xxx}(a, b) + 3(x-a)^2(y-b)f_{xxy}(a, b) + \\ + 3(x-a)(y-b)^2 f_{xyy}(a, b) + (y-b)^3 f_{yyy}(a, b) + \dots] \end{aligned}$$

$$f(x, y) = x^2 y + 3y - 2$$

$$f_x = 2xy$$

$$f_{xx} = 2y, \quad f_{xxx} = 0$$

$$f_y = x^2 + 3$$

$$f_{yy} = 2x, \quad f_{yyy} = 2$$

$$f_{xy} = 2x$$

$$f(1, -2) = -10$$

$$f_x(1, -2) = -4$$

$$f_{xx}(1, -2) = -4, \quad f_{xxx}(1, -2) = 0$$

$$f_y(1, -2) = 4$$

$$f_{yy}(1, -2) = 2, \quad f_{yyy}(1, -2) = 2$$

$$f_{xy}(1, -2) = 2$$

$$f_{xy} = 2$$

$$f_{xy}(1, -2) = 2$$

$$f_{xyy} = 0$$

$$f_{xyy}(1, -2) = 0$$

Substituting all these values in $f(x, y)$, we get

$$\begin{aligned} x^2 y + 3y - 2 &= -10 + [(x-1)(-4) + (y+2)(4)] \\ &\quad + \frac{1}{2} [(x-1)^2(-4) + 2(x-1)(y+2)(2) + (y+2)^2(0)] \\ &\quad + \frac{1}{6} [3(x-1)^2(y+2)(2)] \end{aligned}$$

$$x^2 y + 3y - 2 = -10 - 4(x-1) + 4(y+2) - 2(x-1)^2 + 2(x-1)(y+2) + (x-1)^2(y+2).$$

Example 6: Expand $\frac{(x+h)(y+k)}{x+h+y+k}$ **in powers of** h **and** k **up to the second degree terms.**

Solution:

$$\text{Let } f(x+h, y+k) = \frac{(x+h)(y+k)}{x+h+y+k}$$

$$\therefore f(x, y) = \frac{xy}{x+y}$$

$$f_x = \frac{(x+y)(y) - xy(1)}{(x+y)^2} = \frac{y^2}{(x+y)^2}, \quad f_y = \frac{(x+y)(x) - xy(1)}{(x+y)^2} = \frac{x^2}{(x+y)^2}$$

$$\begin{aligned} f_{xy} &= \frac{(x+y)^2(2y) - y^2 2(x+y)}{(x+y)^4} \\ &= \frac{2(x+y)[2y(x+y) - 2y^2]}{(x+y)^4} = \frac{2xy}{(x+y)^3} \end{aligned}$$

$$f_{xx} = \frac{(x+y)^2(0) - y^2[2(x+y)]}{(x+y)^4} = \frac{-2y^2}{(x+y)^3}$$

$$f_{yy} = \frac{(x+y)^2(0) - x^2[2(x+y)]}{(x+y)^4} = \frac{-2x^3}{(x+y)^3}$$

The Taylor's series expansion for $f(x+h, y+k)$ in powers of h and k is given by

$$\begin{aligned} f(x+h, y+k) &= f(x, y) + [hf_x + kf_y] + \frac{1}{2!} [h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}] + \dots \\ &= \frac{xy}{x+y} + \left[h \left(\frac{y^2}{(x+y)^2} \right) + k \left(\frac{x^2}{(x+y)^2} \right) \right] \\ &\quad + \frac{1}{2} \left[h^2 \left(\frac{-2y^2}{(x+y)^3} \right) + 2hk \left(\frac{2xy}{(x+y)^3} \right) + k^2 \left(\frac{-2x^2}{(x+y)^3} \right) \right] + \dots \\ f(x+h, y+k) &= \frac{xy}{x+y} + \frac{y^2}{(x+y)^2} h + \frac{x^2}{(x+y)^2} k - \frac{y^2}{(x+y)^3} h^2 \\ &\quad + \frac{2xy}{(x+y)^3} hk - \frac{x^2}{(x+y)^3} k^2 + \dots \end{aligned}$$

Exercises

1. Expand $e^x \sin y$ in powers of x and y up to the third degree terms using Taylor's theorem.
2. Expand $\tan^{-1}\left(\frac{y}{x}\right)$ in powers of $(x-1)$ and $(y-1)$ up to the second degree term.
3. Expand $\sin xy$ in powers of $x-1$ and $y-\frac{\pi}{4}$.
4. Using Taylor's series, prove that $\cos(x+y) = 1 - \frac{(x+y)^2}{2!} + \frac{(x+y)^4}{4!} - \dots \infty$
5. Using Taylor's series, prove that $e^y \log(1+x) = x + xy - \frac{x^2}{2} + \frac{x^3}{3} + \frac{xy^2}{2} - \frac{x^2y}{2} + \dots$

6. Using Taylor's series, Prove that $e^x \log(1+y) = y + xy - \frac{y^2}{2} + \frac{x^2 y}{2} - \frac{xy^2}{2} - \frac{y^3}{3} + \dots$
7. Expand x^y at the point (1,1) up to the second degree terms.

Answers

1. $e^x \sin y = y + xy + \frac{x^2 y}{2} - \frac{y^3}{6} + \dots$
2. $\tan^{-1}\left(\frac{y}{x}\right) = \frac{\pi}{4} - \frac{1}{2}(x-1) + \frac{1}{2}(y-1) + \frac{1}{4}(x-1)^2 - \frac{1}{4}(y-1)^2 + \dots$
3. $\sin xy = 1 - \frac{1}{8}\pi^2(x-1)^2 - \frac{1}{2}\pi(x-1)\left(y - \frac{\pi}{2}\right) - \frac{1}{2}\left(y - \frac{\pi}{2}\right)^2 + \dots$
7. $x^y = 1 + (x-1) + (x-1)(y-1) + \frac{1}{2}(x-1)^2 + \dots$

2.11 MAXIMA AND MINIMA OF FUNCTIONS OF TWO VARIABLES

Maximum Value: A function $f(x, y)$ is said to have a maximum value at $x = a$, $y = b$ if

$$f(a, b) > f(a+h, b+k) \text{ for all small values of } h \text{ and } k$$

Minimum Value: A function $f(x, y)$ is said to have a minimum value at $x = a$, $y = b$ if

$$f(a, b) < f(a+h, b+k) \text{ for all small value of } h \text{ and } k.$$

Extremum Value: A maximum or minimum value of a function is said to be extreme or extremum value of a function $f(x, y)$.

To find the extreme values of a function $f(x, y)$.

- (i) Find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.
- (ii) We solve the equations $\frac{\partial f}{\partial x} = 0$ and $\frac{\partial f}{\partial y} = 0$ for x and y .
- (iii) Find $r = \frac{\partial^2 f}{\partial x^2}$, $s = \frac{\partial^2 f}{\partial x \partial y}$, $t = \frac{\partial^2 f}{\partial y^2}$
- (a) If $rt - s^2 > 0$ and $r < 0$, then $f(x, y)$ has a maximum value at (a, b) .

- (b) If $rt - s^2 > 0$ and $r > 0$, then $f(x, y)$ has a minimum value at (a, b) .
- (c) If $rt - s^2 < 0$, then $f(x, y)$ has no extremum value at (a, b) and the point (a, b) is called a saddle point of the function $f(x, y)$.
- (d) If $rt - s^2 = 0$, this case is not considered for further investigation.

Note: A function $f(x, y)$ is said to be stationary at (a, b) if $\left(\frac{\partial f}{\partial x}\right)_{(a,b)} = 0$ and $\left(\frac{\partial f}{\partial y}\right)_{(a,b)} = 0$.

Every extreme value is a stationary value however the converse need not be true.

Geometrical Meaning: The surface $z = f(x, y)$ has a maximum value at the point (a, b) if $f(a, b)$ is the greatest in a small neighborhood of the point (a, b) . As such the surface descends in all directions at this point. The point of maximum can be compared with the highest point of a dome.

The surface has a minimum value at (a, b) if $f(a, b)$ is the smallest in a small neighborhood of the point (a, b) . As such the surface ascends in all directions at that point. The point of minimum can be compared with the lowest point of a bowl.

Saddle Point

The tangent plane is horizontal and the surface descends in certain directions and ascends in other directions at the point (a, b) . Such a point is called a saddle point.

Theorem: The necessary conditions for a function $f(x, y)$ to have an extremum at (a, b) are $f_x(a, b) = 0$ or $f_y(a, b) = 0$.

Critical Point: A point (a, b) is said to be a critical point for a function $f(x, y)$ if $f_x(a, b) = 0$ and $f_y(a, b) = 0$

Note: If f has an extremum at (a, b) , then (a, b) is a critical point however the converse is not true.

Example 1: Find the extreme values for $x^2 + y^2 + 6x + 12$.

Solution:

Let $f(x, y) = x^2 + y^2 + 6x + 12$

$$\frac{\partial f}{\partial x} = 2x + 6 \quad ; \quad \frac{\partial f}{\partial y} = 2y$$

$$r = \frac{\partial^2 f}{\partial x^2} = 2 \quad ; \quad s = \frac{\partial^2 f}{\partial x \partial y} = 0 \quad ; \quad t = \frac{\partial^2 f}{\partial y^2} = 2$$

For extremum, $\frac{\partial f}{\partial x} = 0$ and $\frac{\partial f}{\partial y} = 0$

$$2x + 6 = 0 \text{ and } 2y = 0$$

On solving we get $x = -3$ and $y = 0$.

$$\text{At } (-3, 0), \quad rt - s^2 = 4 - 0 = 4$$

$$rt - s^2 = 4 > 0 \text{ and } r = 2 > 0.$$

$\therefore f(x, y)$ is minimum at $(-3, 0)$.

The minimum value of $f(-3, 0)$ is given by $(-3)^2 + 0 + 6(-3) + 12 = 3$.

Example 2: Find the maximum and minimum of $x^3 + y^3 - 3xy$.

Solution:

$$f(x, y) = x^3 + y^3 - 3xy.$$

$$\frac{\partial f}{\partial x} = 3x^2 - 3y \quad ; \quad \frac{\partial f}{\partial y} = 3y^2 - 3x$$

$$s = \frac{\partial^2 f}{\partial x \partial y} = -3 \quad ; \quad r = \frac{\partial^2 f}{\partial x^2} = 6x \quad ; \quad t = \frac{\partial^2 f}{\partial y^2} = 6y$$

For extremum $\frac{\partial f}{\partial x} = 0$ and $\frac{\partial f}{\partial y} = 0$

$$3x^2 - 3y = 0 \text{ and } 3y^2 - 3x = 0$$

$$x^2 - y = 0 \quad \dots (1) \quad \text{and} \quad y^2 - x = 0 \quad \dots (2)$$

Sub $y = x^2$ in equation (2) we get, $x^4 - x = 0$

$$x(x^3 - 1) = 0$$

$$x(x-1)(x^2 + x + 1) = 0.$$

∴ The real roots are $x = 0, 1$.

∴ $y = 0, 1$.

Hence (0,0) and (1,1) are the stationary points.

At (0,0) , $rt - s^2 = -9 < 0$

At (1,1) , $rt - s^2 = 27 > 0$ and also $r > 0$.

Therefore, (1,1) is a point of minimum value. The minimum value of $f(1,1) = 1 + 1 - 3(1)(1) = -1$.

Example 3: Find the extreme values of $f(x, y) = x^4 + y^4 - 2x^2 + 4xy - 2y^2$.

Solution:

$$f(x, y) = x^4 + y^4 - 2x^2 + 4xy - 2y^2.$$

$$\frac{\partial f}{\partial x} = 4x^3 - 4x + 4y, \quad \frac{\partial f}{\partial y} = 4y^3 + 4x - 4y$$

$$r = \frac{\partial^2 f}{\partial x^2} = 12x^2 - 4, \quad s = \frac{\partial^2 f}{\partial x \partial y} = 4, \quad t = \frac{\partial^2 f}{\partial y^2} = 12y - 4$$

For extremum, $\frac{\partial f}{\partial x} = 0$ and $\frac{\partial f}{\partial y} = 0$.

$$4x^3 - 4x + 4y = 0 \quad \dots (1)$$

$$4y^3 + 4x - 4y = 0 \quad \dots (2)$$

Adding (1) and (2) we get

$$4x^3 + 4y^3 = 0$$

$$x^3 + y^3 = 0, (x+y)(x^2 + xy + y^2) = 0$$

∴ $x = -y$.

Sub $x = -y$ in equation (2) we get, $4y^3 - 8y = 0$, $y^3 - 2y = 0$, $y(y^2 - 2) = 0$

$$\therefore y = 0, \pm \sqrt{2}$$

Substituting $y = 0, \pm \sqrt{2}$ in $x = -y$ we get $x = 0, x = -\sqrt{2}, +\sqrt{2}$.

Hence the stationary points are $(0,0)$, $(-\sqrt{2}, \sqrt{2})$ and $(\sqrt{2}, -\sqrt{2})$.

At $(-\sqrt{2}, \sqrt{2})$, $rt - s^2 = 384 > 0$ and $r > 0$.

At $(\sqrt{2}, -\sqrt{2})$, $rt - s^2 = 384 > 0$ and $r > 0$.

At $(0,0)$, $rt - s^2 = 0$

At $(-\sqrt{2}, \sqrt{2})$ and $(\sqrt{2}, -\sqrt{2})$ the function $f(x, y)$ has a minimum value.

$\therefore$ The minimum value is 8 at $(-\sqrt{2}, \sqrt{2})$.

Example 4: Find the minimum value of $x^2 + y^2 + z^2$ when $x + y + z = 3a$.

Solution:

To find the minimum of $f(x, y) = x^2 + y^2 + z^2$ subject to the constraint $x + y + z = 3a$.

Also $z = 3a - x - y$.

$$f(x, y) = x^2 + y^2 + (3a - x - y)^2$$

$$\frac{\partial f}{\partial x} = 2x + 2(3a - x - y)(-1) = 2x - 2(3a - x - y)$$

$$\frac{\partial f}{\partial y} = 2y + 2(3a - x - y)(-1) = 2y - 2(3a - x - y)$$

$$r = \frac{\partial^2 f}{\partial x^2} = 2 - 2(-1) = 4, \quad s = \frac{\partial^2 f}{\partial x \partial y} = 0 - 2(-1) = 2, \quad t = \frac{\partial^2 f}{\partial y^2} = 2 - 2(-1) = 4$$

For extremum, $\frac{\partial f}{\partial x} = 0$, $\frac{\partial f}{\partial y} = 0$

$$2x + y = 3a, \quad x + 2y = 3a$$

Solving these we get only one extreme points (a, a)

At (a, a) , $rt - s^2 > 0$ and $r > 0$.

$\therefore f(x, y)$ is minimum at (a, a) and the minimum value of $f(a, a) = 3a^2$.

Example 5: Find the points on $z^2 = xy + 1$ which is nearest to the origin.

Solution:

If the required point on the surface is (x, y, z) then its distance from the origin must be minimum.

Hence, we want to find the minimum of $(x-0)^2 + (y-0)^2 + (z-0)^2 = x^2 + y^2 + z^2$.

Let $f(x, y, z) = x^2 + y^2 + z^2$

Using $z^2 = xy + 1$

$f(x, y) = x^2 + y^2 + xy + 1$

$$\frac{\partial f}{\partial x} = 2x + y \quad \dots (1) \quad ; \quad \frac{\partial f}{\partial y} = 2y + x \quad \dots (2)$$

$$r = 2, \quad s = 1, \quad t = 2$$

$$rt - s^2 = 4 - 1 = 3 > 0 \text{ at all points.}$$

Solving equations (1) & (2) we get $x = 0$ and $y = 0$.

Substituting $x = 0$ and $y = 0$ in $z^2 = xy + 1$, we get $z = \pm 1$.

Hence the stationary points are $(0, 0, 1)$ and $(0, 0, -1)$.

Since $rt - s^2 = 3 > 0$ and $r > 0$.

$\therefore f$ is minimum at $(0, 0, 1)$ and $(0, 0, -1)$.

Hence the two nearest points are $(0, 0, 1)$ and $(0, 0, -1)$.

Example 6: Find the maximum and minimum of $x^3 y^2 (1 - x - y)$.

Solution:

$$\text{Let } f(x, y) = x^3 y^2 (1 - x - y)$$

$$\frac{\partial f}{\partial x} = (1 - x - y) 3x^2 y^2 + x^3 y^2 (-1) = (1 - x - y) 3x^2 y^2 - x^3 y^2$$

$$\frac{\partial f}{\partial y} = (1 - x - y) 2x^3 y + x^3 y^2 (-1) = 2x^3 y (1 - x - y) - x^3 y^2$$

$$\begin{aligned} r = \frac{\partial^2 f}{\partial x^2} &= (1 - x - y) 6xy^2 + 3x^2 y^2 (-1) - 3x^2 y^2 \\ &= (1 - x - y) 6xy^2 - 6x^2 y^2 \end{aligned}$$

$$\begin{aligned} t = \frac{\partial^2 f}{\partial y^2} &= (1 - x - y) 2x^3 + 2x^3 y (-1) - 2x^3 y \\ &= (1 - x - y) 2x^3 - 4x^3 y \end{aligned}$$

$$\begin{aligned} s = \frac{\partial^2 f}{\partial x \partial y} &= (1 - x - y) 6x^2 y + 2x^3 y (-1) - 3x^2 y^2 \\ &= (1 - x - y) 6x^2 y - 2x^3 y - 3x^2 y^2. \end{aligned}$$

$$\text{For extremum, } \frac{\partial f}{\partial x} = 0 \Rightarrow (1 - x - y) 3x^2 y^2 - x^3 y^2 = 0$$

$$x^2 y^2 [(1 - x - y) 3 - x] = 0$$

$$3 - 3x - 3y - x = 0, \quad -4x - 3y = -3$$

$$4x + 3y = 3$$

... (1)

$$\frac{\partial f}{\partial y} = 0 \Rightarrow (1 - x - y) 2x^3 y - x^3 y^2 = 0$$

$$x^3 y [2(1 - x - y) - y] = 0, \quad 2 - 2x - 2y - y = 0$$

$$-2x - 3y = -2$$

$$2x + 3y = 2$$

...(2)

Solving equations (1) and (2) we get, $x = \frac{1}{2}, y = \frac{1}{3}$

$\therefore$ Substituting $\left(\frac{1}{2}, \frac{1}{3}\right)$, we get $r = \frac{-1}{9}, s = \frac{-1}{12}, t = \frac{-1}{8}$.

$$\therefore rt - s^2 = \frac{1}{72} - \frac{1}{144} = > 0 \text{ and } r < 0$$

$\therefore f(x, y)$ is maximum at $\left(\frac{1}{2}, \frac{1}{3}\right)$.

Hence the maximum value of $f\left(\frac{1}{2}, \frac{1}{3}\right) = \frac{1}{432}$.

Example 7: Find the maximum and minimum values of

$$f(x, y) = \sin x \sin y \sin(x + y), \quad 0 < x, y < \pi.$$

Solution:

$$f(x, y) = \sin x \sin y \sin(x + y)$$

$$\frac{\partial f}{\partial x} = \cos x \sin y \sin(x + y) + \sin x \sin y \cos(x + y)$$

$$\frac{\partial f}{\partial y} = \sin x \cos y \sin(x + y) + \sin x \sin y \cos(x + y)$$

$$\frac{\partial f}{\partial x} = \sin y \{ \sin(x + y) \cos x + \cos(x + y) \sin x \} = \sin y \sin(2x + y)$$

$$\frac{\partial f}{\partial y} = \sin x \{ \sin(x + y) \cos y + \cos(x + y) \sin y \} = \sin x \sin(x + 2y)$$

$$\frac{\partial^2 f}{\partial x^2} = 2 \sin y \cos(2x + y), \quad \frac{\partial^2 f}{\partial y^2} = 2 \sin x \cos(x + 2y)$$

$$\frac{\partial^2 f}{\partial x \partial y} = \sin y \cos(2x + y) + \sin(2x + y) \cos y = \sin(2x + 2y)$$

For extremum value of $f(x, y)$, $\frac{\partial f}{\partial x} = 0$

$$\text{(i.e.) } \sin y \sin(2x + y) = 0$$

$$\frac{1}{2} [\cos 2x - \cos(2x + 2y)] = 0$$

$$\cos 2x - \cos(2x + 2y) = 0 \quad \dots(1)$$

$$\frac{\partial f}{\partial y} = 0 \quad \text{(i.e.), } \sin x \sin(x + 2y) = 0$$

$$\frac{1}{2} [\cos 2y - \cos(2x + 2y)] = 0$$

$$\cos 2y - \cos(2x + 2y) = 0 \quad \dots(2)$$

From equations (1) and (2) we get $\cos 2x = \cos 2y$

$$\Rightarrow x = y$$

Substituting $x = y$ in equation (1) we get $\cos 2x - \cos 4x = 0$

$$\text{(i.e.) } 2 \sin x \sin 3x = 0$$

$$\therefore \sin x = 0 \text{ or } \sin 3x = 0$$

$$\therefore x = 0, \pi \text{ and } 3x = 0, \pi, 2\pi$$

$$x = 0, \frac{\pi}{3}, \frac{2\pi}{3}$$

$$\therefore \text{The admissible values are } x = 0, \frac{\pi}{3}, \frac{2\pi}{3}.$$

Thus the maxima and minima of $f(x, y)$ are given by $(0, 0)$, $\left(\frac{\pi}{3}, \frac{\pi}{3}\right)$ and $\left(\frac{2\pi}{3}, \frac{2\pi}{3}\right)$.

At the point $(0, 0)$, $r = t = s = 0$.

$$rt - s^2 = 0.$$

Thus the extremum of $f(x, y)$ at $(0, 0)$ is unpredicted.

At the point $\left(\frac{\pi}{3}, \frac{\pi}{3}\right)$, $r = -\sqrt{3}$, $s = -\frac{\sqrt{3}}{2}$, $t = -\sqrt{3}$ and $rt - s^2 = 3 - \frac{3}{4} > 0$

As $rt - s^2 > 0$ and $r < 0$.

$\therefore f(x, y)$ is maximum at $\left(\frac{\pi}{3}, \frac{\pi}{3}\right)$

Maximum value of $f(x, y) = \frac{3\sqrt{3}}{8}$.

At the point $\left(\frac{2\pi}{3}, \frac{2\pi}{3}\right)$, $r = \sqrt{3}$, $s = \frac{\sqrt{3}}{2}$, $t = \sqrt{3}$

$$rt - s^2 = 3 - \frac{3}{4} > 0$$

As $rt - s^2 > 0$ as $r > 0$

$\therefore f(x, y)$ is minimum at $\left(\frac{2\pi}{3}, \frac{2\pi}{3}\right)$

$\therefore$ Minimum value of $f(x, y) = -\frac{3\sqrt{3}}{8}$.

Example 8: In a plane triangle ABC , find the maximum value of $\cos A \cos B \cos C$.

Solution:

$$\text{Let } f = \cos A \cos B \cos C$$

In a triangle ABC , $A + B + C = \pi$, $C = \pi - (A + B)$

$$f = \cos A \cos B \cos[\pi - (A + B)]$$

$$f = \cos A \cos B [-\cos(A + B)]$$

$$f = -\cos A \cos B \cos(A + B)$$

$$\frac{\partial f}{\partial A} = -\cos B [-\sin A \cos(A + B) - \cos A \sin(A + B)]$$

$$= \cos B \sin[A + (A + B)]$$

$$= \cos B \sin(2A + B)$$

$$\frac{\partial f}{\partial B} = -\cos A [-\sin B \cos(A + B) - \cos B \sin(A + B)]$$

$$= \cos A \sin[B + (A + B)]$$

$$= \cos A \sin(A + 2B)$$

$$r = \frac{\partial^2 f}{\partial A^2} = 2 \cos B \sin(2A + B)$$

$$s = \frac{\partial^2 f}{\partial A \partial B} = -\sin A \sin(A + 2B) + \cos A \cos(A + 2B)$$

$$= \cos[2A + 2B]$$

$$t = \frac{\partial^2 f}{\partial B^2} = 2 \cos A \cos(A + 2B)$$

Now, the extremum are given by $\frac{\partial f}{\partial A} = 0$ and $\frac{\partial f}{\partial B} = 0$

$$\Rightarrow \cos B \sin(2A + B) = 0 \quad \dots(1)$$

$$\cos A \sin(A + 2B) = 0 \quad \dots(2)$$

$$\therefore B = 90^\circ \text{ (or) } 2A + B = 180^\circ \text{ and } A = 90^\circ \text{ (or) } 2B + A = 180^\circ$$

The cases $A = 90^\circ$ and $B = 90^\circ$ are impossible.

$$\therefore \text{Only possibility is } 2A + B = 180^\circ \text{ and } 2B + A = 180^\circ.$$

Solving for A and B , we get

$$A = B = 60^\circ \Rightarrow C = \pi - (A + B)$$

$$\text{Hence } C = 60^\circ$$

$$\text{At } A = B = C = 60^\circ,$$

$$r = \frac{\partial^2 f}{\partial A^2} = 2 \cos 60^\circ \cos 180^\circ = -1$$

$$s = \frac{\partial^2 f}{\partial A \partial B} = \cos 240^\circ = \frac{-1}{2}$$

$$t = \frac{\partial^2 f}{\partial B^2} = 2 \cos 60^\circ \cos 180^\circ = -1$$

$$\therefore rt - s^2 = 1 - \frac{1}{4} = \frac{3}{4} > 0 \quad \text{and} \quad r = -1 < 0$$

Hence f is maximum at (A, B) ie, $(60^\circ, 60^\circ)$

$$= -\cos 60^\circ \cos 60^\circ \cos(120^\circ) = \frac{1}{8}$$

Exercises

- Find the extreme values for $x^2 - xy + y^2 + 3x - 2y + 1$.
- Find the extreme values for $x^2y + xy^2 - 12xy$.
- Find the maxima and minima for the function $x^3 + y^3 - 63(x + y) + 12xy$.
- Locate the stationary points of $x^4 + y^4 - 2x^2 - 2y^2 + 4xy$ and determine their nature.
- Show that of all triangles inscribed in a circle, the one with maximum area is equilateral.
- Find the maxima and minima for $\sin x + \sin y + \sin(x + y)$.
- Show that the maximum value of $x^m y^n$ subject to $x > 0, y > 0$ and $x + y = a$ is $\frac{m^m n^n a^{m+n}}{(m+n)^{m+n}}$.
- If $u = 400xyz^2$, show that the maximum value of u is 50 subject to $x^2 + y^2 + z^2 = 1$.
- Show that the stationary value of f given by $f = \frac{a^3}{x^2} + \frac{b^3}{y^2} + \frac{c^3}{z^2}$ subject to

$$x + y + z = 1 \text{ occurs at } \left(\frac{a}{a+b+c}, \frac{b}{a+b+c}, \frac{c}{a+b+c} \right).$$

10. Prove that $x^2y^2 - 5x^2 - 8xy - 5y^2$ is maximum at the origin.

11. Prove that $2x^2y + x^2 - y^2 + 2y$ has no extremum.

Answers

1. No extremum exists.

2. Extremum at $(4, 4)$ is -64.

3. Maxima and minima at (a, a) is $-a^3$.

4. Stationary points are $(0, 0), (\sqrt{2}, -\sqrt{2})$ and $(-\sqrt{2}, \sqrt{2})$.

6. Maxima at $(\pi/3, \pi/3)$ is $\frac{3\sqrt{3}}{2}$.

2.11.1 Lagrange's Method of Undetermined Multipliers

Let $f(x, y, z)$ be a function of three variables subject to the constraint

$$\varphi(x, y, z) = 0 \quad \dots(1)$$

The stationary values of $f(x, y, z)$ for the values of x, y, z are given by the equations,

$$\frac{\partial f}{\partial x} + \lambda \frac{\partial \varphi}{\partial x} = 0 \quad \dots(2)$$

$$\frac{\partial f}{\partial y} + \lambda \frac{\partial \varphi}{\partial y} = 0 \quad \dots(3)$$

$$\frac{\partial f}{\partial z} + \lambda \frac{\partial \varphi}{\partial z} = 0 \quad \dots(4)$$

Solving the equations (1), (2), (3) and (4), we get the values of x, y, z, λ which extremes the $f(x, y, z)$ values.

In this case, λ is an unknown multiplier and is known as Lagrange multiplier.

Note: To find the values of x, y, z for which $f(x, y, z)$ can have a conditional extremum and then construct the auxiliary function $F(x, y, z)$ given by

$$F(x, y, z) = f(x, y, z) + \lambda g(x, y, z)$$

Find $\frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}, \frac{\partial F}{\partial z}$ and $\frac{\partial F}{\partial \lambda}$ and then equate to zero.

$$\text{(i.e.) } \frac{\partial F}{\partial x} = 0; \frac{\partial F}{\partial y} = 0; \frac{\partial F}{\partial z} = 0; \frac{\partial F}{\partial \lambda} = 0.$$

From this, we get the values for x, y, z .

Example 1: Prove that the stationary values of $a^3x^2 + b^3y^2 + c^3z^2$ where $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 1$

occurs at $x = \frac{a+b+c}{a}, y = \frac{a+b+c}{b}, z = \frac{a+b+c}{c}$.

Solution:

$$\text{Let } f = a^3x^2 + b^3y^2 + c^3z^2 \text{ and } \phi = \frac{1}{x} + \frac{1}{y} + \frac{1}{z} - 1.$$

$$\therefore g = f + \lambda \phi \Rightarrow g = a^3x^2 + b^3y^2 + c^3z^2 + \lambda \left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \right)$$

$$\frac{\partial g}{\partial x} = 0, \quad \frac{\partial g}{\partial y} = 0, \quad \frac{\partial g}{\partial z} = 0$$

$$2a^3x - \frac{\lambda}{x^2} = 0, \quad 2b^3y - \frac{\lambda}{y^2} = 0, \quad 2c^3z - \frac{\lambda}{z^2} = 0$$

$$2a^3x^3 = \lambda, \quad 2b^3y^3 = \lambda, \quad 2c^3z^3 = \lambda$$

$$\therefore a^3x^3 = b^3y^3 = c^3z^3 = \frac{\lambda}{2}$$

$$ax = by = cz$$

$$(i.e.) \quad \frac{a}{\frac{1}{x}} = \frac{b}{\frac{1}{y}} = \frac{c}{\frac{1}{z}} = \frac{a+b+c}{\frac{1}{x} + \frac{1}{y} + \frac{1}{z}} = \frac{a+b+c}{1}$$

$$\Rightarrow \frac{a}{\frac{1}{x}} = a+b+c, \frac{b}{\frac{1}{y}} = a+b+c, \frac{c}{\frac{1}{z}} = a+b+c$$

$$\therefore x = \frac{a+b+c}{a}, y = \frac{a+b+c}{b}, z = \frac{a+b+c}{c}$$

Example 2: A rectangular box which is open at the top is to have a volume of 32cm^3 .

Find the dimension of the box, that requires the least material for its construction.

Solution:

Let x, y, z be the length, breadth and height of the box respectively.

If the surface area of the box is least, then the materials for the construction will be least.

(i.e.) To minimize, $S = xy + 2yz + 2zx$.

Subject to the condition that the volume of the box (i.e.) $xyz = 32$.

Consider $f = xy + 2yz + 2zx$ and $\phi = xyz - 32$ the auxillary function is given by

$$g = f + \lambda\phi$$

The stationary points of g are

$$\frac{\partial g}{\partial x} = 0, \frac{\partial g}{\partial y} = 0, \frac{\partial g}{\partial z} = 0, \frac{\partial g}{\partial \lambda} = 0$$

$$\therefore g = xy + 2yz + 2zx + \lambda(xyz - 32)$$

$$\frac{\partial g}{\partial x} = y + 2z + \lambda yz = 0 \quad \dots(1)$$

$$\frac{\partial g}{\partial y} = x + 2z + \lambda xz = 0 \quad \dots(2)$$

$$\frac{\partial g}{\partial z} = 2x + 2y + \lambda xy = 0 \quad \dots(3)$$

$$\frac{\partial g}{\partial \lambda} = xyz - 32 = 0 \quad \dots(4)$$

$$\frac{(1)}{yz} \Rightarrow \frac{1}{z} + \frac{2}{y} = -\lambda, \quad \frac{(2)}{xz} \Rightarrow \frac{1}{z} + \frac{2}{x} = -\lambda, \quad \frac{(3)}{xy} \Rightarrow \frac{2}{x} + \frac{2}{z} = -\lambda$$

$$\text{Solving three equations, we get, } x = \frac{-4}{\lambda}, y = \frac{-4}{\lambda}, z = \frac{-2}{\lambda}$$

Using these values in equation (4) we get,

$$\lambda = -1 \text{ is the only possible value.}$$

$$\therefore x = 4, y = 4, z = 2$$

$\therefore$ Thus the dimensions of the box are 4cm, 4cm, 2cm .

Example 3: Find the greatest and the least distances of the point $(1, 2, -2)$ from the unit sphere whose centre is at the origin.

Solution:

The equation of the unit sphere with centre $(0, 0, 0)$ is $x^2 + y^2 + z^2 = 1$.

The distance of the point (x, y, z) from $(1, 2, -2)$ of the sphere is

$$d = \sqrt{(x-1)^2 + (y-2)^2 + (z+2)^2}.$$

$$\therefore f = (x-1)^2 + (y-2)^2 + (z+2)^2 \text{ subject to the constraint } x^2 + y^2 + z^2 - 1 = 0.$$

The auxillary function is given by $g = f + \lambda \phi$

$$g = (x-1)^2 + (y-2)^2 + (z+2)^2 + \lambda (x^2 + y^2 + z^2 - 1)$$

The stationary values are given by

$$\frac{\partial g}{\partial x} = 0, \quad \frac{\partial g}{\partial y} = 0, \quad \frac{\partial g}{\partial z} = 0, \quad \frac{\partial g}{\partial \lambda} = 0$$

$$\frac{\partial g}{\partial x} = 2(x-1) + 2x\lambda = 0 \quad \dots (1)$$

$$x - 1 + x\lambda = 0$$

$$x(1+\lambda)=1$$

$$x = \frac{1}{1+\lambda}$$

$$\frac{\partial g}{\partial y} = 2(y-2) + 2y\lambda = 0 \quad \dots (2)$$

$$y-2+\lambda y=0$$

$$y = \frac{2}{1+\lambda}$$

$$\frac{\partial g}{\partial z} = 2(z+2) + 2z\lambda = 0 \quad \dots (3)$$

$$z+2+\lambda z=0$$

$$z(1+\lambda)=-2$$

$$z = \frac{-2}{1+\lambda}.$$

$$\frac{\partial g}{\partial \lambda} = x^2 + y^2 + z^2 - 1 = 0$$

$$x^2 + y^2 + z^2 = 1 \quad \dots (4)$$

Sub these values in equation (4) we get,

$$\frac{1}{(1+\lambda)^2} + \frac{4}{(1+\lambda)^2} + \frac{4}{(1+\lambda)^2} = 1$$

$$\frac{9}{(1+\lambda)^2} = 1$$

$$(\text{i.e.}) \quad (1+\lambda)^2 = 9$$

$$1+2\lambda+\lambda^2=9$$

$$\lambda^2+2\lambda-8=0$$

$$(\lambda - 2)(\lambda + 4) = 0$$

$$\lambda = -4 \text{ (or) } \lambda = 2$$

When $\lambda = -4$, the point on the sphere is $\frac{-1}{3}, \frac{-2}{3}, \frac{2}{3}$ and its distance is $d = \frac{12}{3}$.

When $\lambda = 2$, the point on the sphere is $\left(\frac{1}{3}, \frac{2}{3}, \frac{-2}{3}\right)$ and its distance is $d = \frac{2}{\sqrt{3}}$.

$\therefore$ The least and the greatest distances are $\frac{12}{3}$ and $\frac{2}{\sqrt{3}}$ respectively.

Example 4: Find the volume of the greatest rectangular parallelepiped which can be

inscribed in the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.

Solution:

Let the vertex of rectangular parallelepiped be (x, y, z) , then the edges of the parallelepiped be $2x, 2y, 2z$. (lengths).

$$\therefore \text{Volume (V)} = 2x \cdot 2y \cdot 2z = 8xyz = f \text{ (say)} \quad \dots (1)$$

$$\therefore \text{We have to maximize V subject to the condition } \varphi = \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1 = 0 \quad \dots (2)$$

The auxiliary function F is given by $F = f + \lambda \varphi$

$$F = 8xyz + \lambda \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1 \right)$$

$$\frac{\partial F}{\partial x} = 0 \Rightarrow 8yz + \frac{2x\lambda}{a^2} = 0$$

$$\frac{\partial F}{\partial y} = 0 \Rightarrow 8xz + \frac{2y\lambda}{b^2} = 0$$

$$\frac{\partial F}{\partial z} = 0 \Rightarrow 8xy + \frac{2z\lambda}{c^2} = 0$$

From the above equations, we get $\lambda = \frac{-4a^2yz}{x} = \frac{-4b^2xz}{y} = \frac{-4c^2xy}{z}$

$$\therefore \frac{a^2xyz}{x^2} = \frac{b^2xyz}{y^2} = \frac{c^2xyz}{z^2}$$

(i.e.) Dividing by xyz and rewriting we get

$$\frac{\frac{x^2}{a^2}}{1} = \frac{\frac{y^2}{b^2}}{1} = \frac{\frac{z^2}{c^2}}{1} = \frac{\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2}}{3} = \frac{1}{3} \quad (\because \text{using equation (2)})$$

$$x^2 = \frac{a^2}{3} \Rightarrow x = \frac{a}{\sqrt{3}}$$

$$y^2 = \frac{b^2}{3} \Rightarrow y = \frac{b}{\sqrt{3}}$$

$$z^2 = \frac{c^2}{3} \Rightarrow z = \frac{c}{\sqrt{3}}$$

$\therefore$ The extremum point is $\left(\frac{a}{\sqrt{3}}, \frac{b}{\sqrt{3}}, \frac{c}{\sqrt{3}} \right)$

The maximum value of $V = 8xyz$ at the extremum point is $\left(\frac{8abc}{3\sqrt{3}} \right)$.

Example 5: Find the shortest distance of the focus $(1, 0)$ from any point of the parabola

$$y^2 - 4x = 0.$$

Solution:

Let (x, y) be the point nearest to $(1, 0)$.

Then distance $d = \sqrt{(x-1)^2 + y^2}$

$$f = (x-1)^2 + y^2 (= d^2)$$

We have to minimize f subject to $\varphi = y^2 - 4x = 0$.

Extreme points are given by

$$\frac{\partial f}{\partial x} + \lambda \frac{\partial \varphi}{\partial x} = 0 \Rightarrow 2(x-1) - 4\lambda = 0 \quad \dots (1)$$

$$\frac{\partial f}{\partial y} + \lambda \frac{\partial \varphi}{\partial y} = 0 \Rightarrow 2y + \lambda 2y = 0 \Rightarrow y(1 + \lambda) = 0 \quad \dots (2)$$

Solving equation (2), we get $y = 0$ (or) $\lambda = -1$.

Substituting $\lambda = -1$ in equations (1), we get $x = -1 \Rightarrow y^2 - 4x = 0$ becomes imaginary.

Take $y = 0$, then $y^2 - 4x = 0 \Rightarrow x = 0$.

$\therefore (0, 0)$ gives the extremum points.

At $(0, 0)$, extremum value is $f = (0-1)^2 + 0 = 1$.

$$\therefore d = \sqrt{f} = 1$$

This gives the shortest distance of the focus $(1, 0)$ to the parabola $y^2 - 4x = 0$.

Example 6: Show that the greatest rectangle that can be inscribed in the ellipse $4x^2 + 9y^2 = 36$, having its sides parallel to the axes is 12.

Solution:

Let the vertices of the rectangle be $A(x, y), B(-x, y), C(-x, -y), D(x, -y)$ so that A, B, C, D lies on the ellipse $4x^2 + 9y^2 = 36$.

$$\begin{aligned} \text{Area of the rectangle, } A &= lb = (2x)(2y) \\ &= 4xy = f \text{ (say).} \end{aligned}$$

We have to maximize the area of the rectangle A subject to the condition,

$$\varphi = 4x^2 + 9y^2 - 36 = 0 \quad \dots (1)$$

The auxiliary function F is given by $F(x, y) = f + \lambda \varphi$

$$F = 4xy + \lambda(4x^2 + 9y^2 - 36)$$

$$\frac{\partial F}{\partial x} = 4y + \lambda(8x) = 0 \quad \dots (2) \quad ; \quad \frac{\partial F}{\partial y} = 4x + \lambda(18y) = 0 \quad \dots (3)$$

$$\frac{\partial F}{\partial \lambda} = 0 + (4x^2 + 9y^2 - 36)(1) = 0 \quad \dots (4)$$

From equations (3) and (4) we get,

$$16xy = 144\lambda^2 xy$$

$$\lambda^2 = \frac{1}{9} \Rightarrow \lambda = \pm \frac{1}{3}$$

$$\text{Sub } \lambda = -\frac{1}{3} \text{ in equation (4) we get } 4x - 6y = 0 \quad ; \quad 2x = 3y \quad \dots (5)$$

Substituting equation (5) in (1) we get $9y^2 + 9y^2 - 36 = 0$

$$18y^2 = 36$$

$$y^2 = 2 \Rightarrow y = \pm\sqrt{2}$$

$$\therefore \text{Substituting } y = \pm\sqrt{2} \text{ in equation (5) we get, } 2x = \pm 3\sqrt{2} \Rightarrow x = \frac{\pm 3\sqrt{2}}{2}$$

$\therefore$ The vertices of the rectangle are

$$A\left(\frac{3\sqrt{2}}{2}, \sqrt{2}\right), B\left(-\frac{3\sqrt{2}}{2}, \sqrt{2}\right), C\left(-\frac{3\sqrt{2}}{2}, -\sqrt{2}\right) \text{ and } D\left(\frac{3\sqrt{2}}{2}, -\sqrt{2}\right).$$

Hence the maximum area of the rectangle is given by A (i.e.) $x = \frac{3\sqrt{2}}{2}, y = \sqrt{2}$.

$\therefore$ The maximum area is $A = 12$.

Example 7: Prove that if the perimeter of a triangle is constant, its area is maximum when the triangle is equilateral.

Solution:

Let a, b, c be the sides of a triangle whose perimeter $2s$ is constant.

Then, $a + b + c = 2s$.

Area of the triangle is $\Delta = \sqrt{s(s-a)(s-b)(s-c)}$

If Δ is maximum, so will be Δ^2 .

Now we have to maximize $f = \Delta^2 = s(s-a)(s-b)(s-c)$ subject to the condition,

$$\phi = a + b + c - 2s = 0 \quad \dots (1)$$

Lagrange's equations are

$$\frac{\partial f}{\partial a} + \lambda \frac{\partial \phi}{\partial a} = -s(s-b)(s-c) + \lambda = 0 \quad \dots (2)$$

$$\frac{\partial f}{\partial b} + \lambda \frac{\partial \phi}{\partial b} = -s(s-a)(s-c) + \lambda = 0 \quad \dots (3)$$

$$\frac{\partial f}{\partial c} + \lambda \frac{\partial \phi}{\partial c} = -s(s-a)(s-b) + \lambda = 0 \quad \dots (4)$$

From equations (2) and (3),

$$\lambda = s(s-a)(s-c) = s(s-c)(s-b)$$

$$s-a = s-b \Rightarrow a = b$$

Similarly from equations (3) and (4), $b = c$

$\therefore \Delta$ has maximum value when $a = b = c$, i.e., when the triangle is equilateral.

Exercises

1. Find the minimum value of $x^2 + y^2$ subject to the condition $ax + by = c$.
2. Show that the rectangular solid of maximum volume that can be inscribed in a sphere is a cube.
3. Find the dimensions of the rectangular box without a top of maximum capacity, whose surface is 108 sq.cm .
4. Find the point on the surface $z^2 = xy + 1$ nearest to the origin.
5. Find the stationary values of $u = x + y + z$ subject to the condition $\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 1$.
6. Find the maximum value of $x^2 y^2 z^2$ under the condition $x^2 + y^2 + z^2 = a^2$.

7. The temperature T at any point (x, y, z) in space is $T = 400xyz^2$. Find the highest temperature on the surface of the unit sphere $x^2 + y^2 + z^2 = 1$.
8. Find the minimum value of $x^2 + y^2 + z^2$, given that $ax + by + cz = p$.
9. Find the shortest and longest distance from the point $(1, 2, -1)$ to the sphere $x^2 + y^2 + z^2 = 24$.

Answers

1. Minimum at $\left(\frac{ac}{a^2 + b^2}, \frac{bc}{a^2 + b^2} \right)$.
2. length = 6cm , breadth = 6cm , height = 3cm .
4. f is minimum at $(0, 0, 1)$ and $(0, 0, -1)$.
5. $x = \sqrt{a}(\sqrt{a} + \sqrt{b} + \sqrt{c})$; $y = \sqrt{b}(\sqrt{a} + \sqrt{b} + \sqrt{c})$; $z = \sqrt{c}(\sqrt{a} + \sqrt{b} + \sqrt{c})$
6. $\left(\frac{a^2}{3} \right)^3$
7. $T = 50^\circ\text{C}$.
8. Minimum value of $u = \frac{p^2}{a^2 + b^2 + c^2}$.
9. Shortest distance = $\sqrt{14}$; Longest distance = $\sqrt{46}$.

PART A – QUESTIONS AND ANSWERS

1. If $z = 2x^2 + 3xy + y^3$, find $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$.

Given that $z = 2x^2 + 3xy + y^3$

$$\frac{\partial z}{\partial x} = 4x + 3y \quad ; \quad \frac{\partial z}{\partial y} = 3x + 3y^2$$

2. If $x = r \cos \theta, y = r \sin \theta$, find $\frac{\partial x}{\partial r}, \frac{\partial x}{\partial \theta}, \frac{\partial y}{\partial r}, \frac{\partial y}{\partial \theta}$.

Given that $x = r \cos \theta$

$$\frac{\partial x}{\partial r} = \cos \theta, \frac{\partial x}{\partial \theta} = -r \sin \theta$$

$$y = r \sin \theta$$

$$\frac{\partial y}{\partial r} = \sin \theta, \frac{\partial y}{\partial \theta} = r \cos \theta.$$

3. Define a homogeneous function.

A function $f(x, y, z)$ is called a homogeneous function of degree n if

$$f(tx, ty, tz) = t^n f(x, y, z), \text{ } n \text{ need not be an integer.}$$

4. Find the second order partial derivatives of $f(x, y) = ax^2 + 2hxy + by^2$.

Given that $f(x, y) = ax^2 + 2hxy + by^2$

$$\frac{\partial f}{\partial x} = 2ax + 2hy; \quad \frac{\partial f}{\partial y} = 2hx + 2by$$

$$\frac{\partial^2 f}{\partial x^2} = 2a; \quad \frac{\partial^2 f}{\partial x \partial y} = 2h; \quad \frac{\partial^2 f}{\partial y^2} = 2b; \quad \frac{\partial^2 f}{\partial y \partial x} = 2h.$$

5. If $u = \log(x^2 + y^2)$, Prove that $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$.

$$\text{Given that } \frac{\partial u}{\partial x} = \frac{1}{x^2 + y^2} (2x) = \frac{2x}{x^2 + y^2}; \quad \frac{\partial u}{\partial y} = \frac{1}{x^2 + y^2} (2y) = \frac{2y}{x^2 + y^2}$$

$$\frac{\partial^2 u}{\partial y \partial x} = \frac{-2x}{(x^2 + y^2)^2} (2y) = \frac{-4xy}{(x^2 + y^2)^2};$$

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{-2y}{(x^2 + y^2)^2} (2x) = \frac{-4xy}{(x^2 + y^2)^2}$$

$$\therefore \frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$$

6. Find $\frac{dy}{dx}$ if $ax^2 + 2hxy + by^2 - 1$.

Given that $f(x, y) = ax^2 + 2hxy + by^2 - 1$

$$\frac{\partial f}{\partial x} = 2ax + 2hy, \quad \frac{\partial f}{\partial y} = 2hx + 2by$$

$$\frac{dy}{dx} = \frac{f_x}{f_y} = \frac{ax + hy}{hx + by}$$

7. Write some example for homogeneous functions.

$$1. \quad f(x, y, z) = x^3 \sin\left(\frac{x}{y}\right) + y^3 \tan^{-1}\left(\frac{y}{x}\right)$$

This is a homogeneous function of degree 3 in x, y, z .

$$2. \quad f(x, y) = \cos\left(\frac{x}{y}\right)$$

This is a homogeneous function of degree zero in x, y, z

8. If $z = (1 - 2xy + y^2)^{\frac{-1}{2}}$, Prove that $x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y} = y^2 z^3$.

Given that $z = (1 - 2xy + y^2)^{\frac{-1}{2}}$

$$\frac{\partial z}{\partial x} = \frac{-1}{2} (1 - 2xy + y^2)^{\frac{-3}{2}} (-2y) = y (1 - 2xy + y^2)^{\frac{-3}{2}}$$

$$\frac{\partial z}{\partial y} = \frac{-1}{2} (1 - 2xy + y^2)^{\frac{-3}{2}} (-2x + 2y) = (1 - 2xy + y^2)^{\frac{-3}{2}} (x - y)$$

$$x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y} = xy (1 - 2xy + y^2)^{\frac{-3}{2}} - y(x - y) (1 - 2xy + y^2)^{\frac{-1}{2}}$$

$$= xyz^3 - y(x - y)z^3$$

$$\therefore x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y} = y^2 z^3$$

9. Find $\frac{du}{dt}$ if $u = \frac{x}{y}$, $x = e^t$, $y = \log t$.

$$\frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} = \frac{1}{y} e^t + \frac{-x}{y^2} \left(\frac{1}{t} \right)$$

$$\frac{du}{dt} = \frac{e^t}{\log t} - \frac{e^t}{t(\log t)^2}$$

10. If $f(x, y) = 0$, $\phi(y, z) = 0$, Prove that $\frac{\partial f}{\partial y} \cdot \frac{\partial \phi}{\partial z} \cdot \frac{dz}{dx} = \frac{\partial f}{\partial x} \cdot \frac{\partial \phi}{\partial y}$.

Given that $f(x, y) = 0$ $\phi(y, z) = 0$

$$\frac{dy}{dx} = \frac{f_x}{f_y} \quad \frac{dz}{dy} = \frac{\phi_y}{\phi_z}$$

$$\frac{dz}{dx} = \frac{dy}{dx} \cdot \frac{dz}{dy} = \frac{f_x}{f_y} \cdot \frac{\phi_y}{\phi_z}$$

$$\begin{aligned} &= \frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}} \cdot \frac{\frac{\partial \phi}{\partial y}}{\frac{\partial \phi}{\partial z}} \\ &= \frac{\partial f}{\partial x} \cdot \frac{\partial \phi}{\partial y} \cdot \frac{\partial z}{\partial x} \end{aligned}$$

$$\therefore \frac{\partial f}{\partial y} \cdot \frac{\partial \phi}{\partial z} \cdot \frac{dz}{dx} = \frac{\partial f}{\partial x} \cdot \frac{\partial \phi}{\partial y}$$

11. What is meant by total differentiation?

In the differential form, $dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$, this dz is called the total differential of z ,

where z is a function of two variables x and y .

12. Define Euler's theorem on homogeneous functions.

If $z = f(x, y)$ is a homogeneous function of x and y of order n , then

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = nf$$

13. Verify Euler's theorem for $f(x, y) = ax^2 + 2hxy + by^2$.

Given that $f(x, y) = ax^2 + 2hxy + by^2$

$$\frac{\partial f}{\partial x} = 2ax + 2hy; \quad \frac{\partial f}{\partial y} = 2hx + 2by$$

$$\begin{aligned} x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} &= x(2ax + 2hy) + y(2hx + 2by) \\ &= 2(ax^2 + 2hxy + by^2) = 2f(x, y) \end{aligned}$$

$f(x, y)$ is homogeneous of degree 2. Hence verified.

14. If $z = \log(x^3 - x^2y - xy^2 + y^3)$ then find $\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}$.

$$\text{Given that } \frac{\partial z}{\partial x} = \frac{1}{(x^3 - x^2y - xy^2 + y^3)} (3x^2 - 2xy - y^2)$$

$$\frac{\partial z}{\partial y} = \frac{1}{(x^3 - x^2y - xy^2 + y^3)} (-x^2 - 2xy + 3y^2)$$

$$\begin{aligned} \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} &= \frac{1}{(x^3 - x^2y - xy^2 + y^3)} (3x^2 - 2xy - y^2 - x^2 - 2xy + 3y^2) \\ &= \frac{1}{(x^3 - x^2y - xy^2 + y^3)} (2x^2 - 4xy + 2y^2) = \frac{2(x-y)^2}{(x^2 - y^2)(x-y)} \end{aligned}$$

$$\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = \frac{2}{x+y}$$

15. If $u = \sin^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{x}{y}\right)$. Find $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$.

Since u is a homogeneous function of degree zero. By Euler's

theorem, $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 0(u) = 0$

16. If $z = f\left(\frac{y}{x}\right) + \phi\left(\frac{x}{y}\right)$, prove that $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 0$.

Given that $z = F(x, y) = f\left(\frac{y}{x}\right) + \phi\left(\frac{x}{y}\right)$

$$F(tx, ty) = f\left(\frac{ty}{tx}\right) + \phi\left(\frac{tx}{ty}\right) = F(x, y)$$

Hence z is homogeneous of degree zero.

By Euler's theorem, $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 0(z) = 0$.

17. If $f(x, y) = x^2 \tan^{-1}\left(\frac{y}{x}\right) - y^2 \sin^{-1}\left(\frac{x}{y}\right)$, Prove that $x^2 \frac{\partial^2 f}{\partial x^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} + y^2 \frac{\partial^2 f}{\partial y^2} = 2f$.

To test the homogeneity of the function,

$$f(tx, ty) = (tx)^2 \tan^{-1}\left(\frac{ty}{tx}\right) - (ty)^2 \sin^{-1}\left(\frac{tx}{ty}\right) = t^2 f(x, y).$$

Hence f is a homogeneous function of x, y of degree 2.

$$\begin{aligned} \therefore x^2 \frac{\partial^2 f}{\partial x^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} + y^2 \frac{\partial^2 f}{\partial y^2} &= n(n-1)f \\ &= 2(2-1)f = 2f. \end{aligned}$$

18. If $f(x, y) = \frac{x^2 + y^2}{x - y}$, Prove that $x^2 \frac{\partial^2 f}{\partial x^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} + y^2 \frac{\partial^2 f}{\partial y^2} = 0$.

$f(x, y)$ is a homogeneous of degree zero.

By extended Euler's theorem, $x^2 \frac{\partial^2 f}{\partial x^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} + y^2 \frac{\partial^2 f}{\partial y^2} = n(n-1)f$

$$\therefore x^2 \frac{\partial^2 f}{\partial x^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} + y^2 \frac{\partial^2 f}{\partial y^2} = 0(f) = 0.$$

19. If $u = xy + yz + zx$ where $x = \frac{1}{t}, y = e^t, z = e^{-t}$, find $\frac{du}{dt}$.

$$\begin{aligned} \frac{du}{dt} &= \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial u}{\partial z} \cdot \frac{dz}{dt} \\ &= (y+z) \left(\frac{-1}{t^2} \right) + (x+z) e^t + (x+y)(-e^{-t}) \\ \frac{du}{dt} &= (x+z)e^t - (x+y)e^{-t} - \left(\frac{y+z}{t^2} \right). \end{aligned}$$

20. If $u = e^x \sin y$ where $x = st^2, y = s^2t$. Find $\frac{\partial u}{\partial s}, \frac{\partial u}{\partial t}$.

$$\begin{aligned} \frac{\partial u}{\partial s} &= \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial s} \\ &= e^x \sin y(t^2) + e^x \cos y(2st) \\ \frac{\partial u}{\partial s} &= e^x [t^2 \sin y + 2st \cos y] \\ \frac{\partial u}{\partial t} &= \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial t} \\ \frac{\partial u}{\partial t} &= e^x [2st \sin y + s^2 \cos y] \end{aligned}$$

21. If $x = r \cos \theta, y = r \sin \theta$, find $\frac{\partial(x, y)}{\partial(r, \theta)}$

$$\text{We know that } \frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix}$$

$$= r \cos^2 \theta + r \sin^2 \theta$$

$$\frac{\partial(x, y)}{\partial(r, \theta)} = r$$

22. Define stationary point.

The point (a, b) is called a stationary point if $f_x(a, b) = 0$ and $f_y(a, b) = 0$. The value of $f(a, b)$ is called the stationary value.

23. If $u = x(1 - y)$, $v = xy$, then the value of the Jacobian $\frac{\partial(u, v)}{\partial(x, y)}$

$$\text{We know that } \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} 1-y & -x \\ y & x \end{vmatrix} = x(1-y) + xy = x$$

$$\frac{\partial(u, v)}{\partial(x, y)} = x$$

24. Define saddle point.

If $\frac{\partial^2 f}{\partial x^2} \frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y} \right)^2 < 0$, then we have neither a maximum nor a minimum, such a point is called a saddle point.

25. Define Lagrange's multiplier.

If $f(x, y, z)$ is a function of the three variables subject to the constraint, $\phi(x, y, z) = 0$ the values of x, y, z that give the stationary values of $f(x, y, z)$ are given by the

$$\text{equations } \frac{\partial f}{\partial x} + \lambda \frac{\partial \phi}{\partial x} = 0$$

$$\frac{\partial f}{\partial y} + \lambda \frac{\partial \phi}{\partial y} = 0$$

$$\frac{\partial f}{\partial z} + \lambda \frac{\partial \phi}{\partial z} = 0 \text{ and } \phi(x, y, z) = 0$$

Solving these equations for x, y, z and λ we obtain the co-ordinates x, y, z which maximizes or minimizes $f(x, y, z)$. Here λ is a parameter independent of x, y, z and it is called Lagrange multiplier.